Chapter IX

Ionizational Nonequilibrium

1. INTRODUCTION

In the preceding chapter we considered the transport effects that are associated with translational nonequilibrium in a partially ionized plasma. There it was implicitly assumed that the electron number density was known or could be independently calculated, as would be the case if n_e were given in terms of T_e by the Saha equation. In addition, it was found that under widely applicable conditions nonelastic collisions do not have a significant direct effect on the calculation of transport properties. As a consequence, the discussion in that chapter emphasized the effects of elastic collisions.

In this chapter, we turn our attention to certain thermodynamic properties of partially ionized plasmas. In particular, we emphasize the interplay between nonelastic electron-atom and electron-ion collisions and radiative processes in establishing the degree of ionization of the plasma and the populations of excited atomic energy levels. Our objective is to discuss some of the more important considerations in this area, rather than to develop a complete and general theory or to obtain numerical formulas that encompass the spectrum of diverse practical applications. In contrast to the calculation of transport properties, the complexity of ionizational nonequilibrium and the dependence of the situation on a set of nonelastic cross sections for the plasma in question makes simple working formulas such as the mixture rules of Chapter VIII, Sec. 7, less readily achieved. Thus, in the present case the approach to a given practical problem, including the applicable approximations as well as the relevant cross sections, will depend on the particular plasma conditions. Nevertheless, we shall be able to indicate at least qualitatively what effects are to be expected and to discuss certain results such as those of Bates, Kingston, and McWhirter

(1962) and the approximate recombination-rate formula of Hinnov and Hirschberg (1962) which have broad applicability.

In accord with the foregoing objective, we restrict our attention to the processes of ionization and excitation of bound electronic levels of monatomic atoms under circumstances for which these processes and their inverses result from either nonelastic collisions of the atoms with free electrons in the plasma or from the emission and absorption of radiation. That is, we do not consider the excitation and ionization of molecular species or the effects of nonelastic atom-atom collisions. In monatomic plasmas the contribution of atom-atom collisions to the processes in question is in fact negligible under many conditions of interest except at low electron number densities. In plasmas where molecular species are present, the exchange of energy between internal molecular degrees of freedom, particularly vibrational energy, and electronic excitation and ionization can be of significant importance. The general approach to such situations and the qualitative effects that occur, however, are similar to those for the simpler monatomic plasmas that we shall discuss.

Furthermore, we shall assume for the most part that as a result of electron-electron collisions the free-electron distribution function is nearly Maxwellian, not only for thermal electrons as in Chapter VIII, Sec. 3, but also for the higher-energy electrons that participate in exciting and ionizing collisions. To do otherwise would excessively complicate an introductory discussion. The more general situation, in which the electron Boltzmann equation for the distribution function is coupled with rate equations of this chapter, is the subject of current research. Such considerations will be discussed in Sec. 5.

In the following section, the continuity and rate equations that govern the electron density and the populations of the various excited electronic levels are derived. These rate equations are applied first, in Sec. 3, to the analysis of a steady-state plasma for which the effects of convection are absent. The influence of the escape of radiation on the level populations and departures from the Saha equation are discussed. We then consider, in Sec. 4, ionization and recombination rates in a transient or flowing plasma. In such cases, the time-derivative or convective terms in the continuity equations for the electrons and the excited atoms must be retained with the pertinent rates given by the rate equations of Sec. 2. Up to this point a Maxwellian distribution for the free electrons is assumed, so that the analysis is uncoupled from the solution of the electron Boltzmann equation. In Sec. 5 we discuss this coupled problem and indicate what modifications result from a non-Maxwellian distribution at low electron densities. In the final section the theory of Sec. 4 is applied to problems of ionizational nonequilibrium in flowing plasmas.

2. CONTINUITY AND RATE EQUATIONS

In a uniform plasma the continuity equation for the population, that is, number density, of atoms in a given bound electronic energy level expresses a balance between the rates of collisional and radiative excitation and de-excitation of the level and the time rate of change of the number density of the level. Similarly, the continuity equation for the free electron density involves the rates of ionization and recombination. In a nonuniform plasma the effects of convection and diffusion must be added to these equations. If, as in Chapter VIII, Sec. 8, n_k denotes the number density of atoms in the kth electronic level and $\dot{n}_k$ the corresponding net volumetric rate of production of such atoms as a result of nonelastic collisional and radiative processes, the continuity equation for the kth level is

$$\frac{\partial n_k}{\partial t} + \nabla \cdot [n_k(\mathbf{u} + \mathbf{U}_k)] = \dot{n}_k. \tag{2.1}$$

As before, $\mathbf{u}$ is the mass velocity of the plasma as a whole and $\mathbf{U}_k$ is the diffusion velocity of atoms in the kth level. The corresponding continuity equation for the electron number density is

$$\frac{\partial n_e}{\partial t} + \nabla \cdot [n_e(\mathbf{u} + \mathbf{U}_e)] = \dot{n}_e. \tag{2.2}$$

A similar equation applies to the ions. These equations can be viewed as derived directly from the physical requirements of continuity for n_k and n_e or, alternatively, as obtained from the integration over velocity space of the excited-atom and electron Boltzmann equations. In the latter interpretation the terms $\dot{n}_k$ and $\dot{n}_e$ represent the velocity integrals of nonelastic-collision terms.

In this section we consider the formulation of the expressions for $\dot{n}_k$ and $\dot{n}_e$. The equations for these quantities are called *rate equations*. For clarity and simplicity we restrict our attention to a quasineutral plasma with a single ionizable species, so that the kth level refers to an electronic level of that species. Furthermore, we neglect multiple ionization and assume that a negligible fraction of the ions are excited above the ionic ground level. Although we use the word level to denote a degenerate set of indistinguishable quantum states of identical energy, in practice the continuity and rate equations are often applied to groups of such levels of nearly the same energy. That is, the kth "level" would then represent several actual levels taken altogether. The utility of this grouping of levels results from the efficiency of collisional equilibration among neighboring levels.

Before discussing the general rate equations, to fix ideas we first consider a useful simplified version of the rate equation for the first excited level.

In many cases, the population n_2 of this level is determined primarily by the rates of collisional and radiative transitions to and from only the ground level ($k = 1$). As indicated previously, we restrict our attention to situations where collisional excitation and de-excitation result from electron-atom collisions. The rate per unit volume of *inelastic excitation collisions* of the type $e + A_1 \rightarrow e + A_2$, where A_1 and A_2 represent atoms in the first and second levels, will be proportional to the product $n_e n_1$, and can be written as

$$n_e n_1 \, {}_1S^2.$$

Under conditions for which the departure of the electron distribution function from an isotropic distribution $f^0(C)$ is small, the *rate integral* ${}_1S^2$ is given in terms of the excitation cross section $Q^{1\rightarrow 2}$ by

$${}_1S^2 = \int_{\sqrt{2\epsilon_{12}/m_e}}^{\infty} f^0(C) C Q^{1\rightarrow 2}(C) 4\pi C^2 \, dC.$$

Here ϵ_{12} is the excitation energy measured from the ground level and $Q^{1\rightarrow 2}$ is the total cross section, integrated over all deflection angles. We have accounted for the consequences of the small electron mass by writing the relative speed g as C and $Q^{1\rightarrow 2}(g)$ as $Q^{1\rightarrow 2}(C)$ and have integrated over all velocities of the atoms. The lower limit on the integration reflects the fact that for excitation the electron kinetic energy $\frac{1}{2}m_e C^2$ must be greater than ϵ_{12}. The cross section $Q^{1\rightarrow 2}$ and the other nonelastic cross sections that enter the present calculations are discussed in Chapter II, where, in Sec. 14, representative values are given.

It is conventional to rewrite the expression for ${}_1S^2$ in terms of the electron energy $\epsilon = \frac{1}{2}m_e C^2$. If we assume that the electron distribution is Maxwellian, that is, that

$$f^0 = f_M(\epsilon) = \left(\frac{m_e}{2\pi kT}\right)^{3/2} e^{-\epsilon/kT_e},$$

we then have

$${}_1S^2 = \int_{\epsilon_{12}}^{\infty} f_M(\epsilon) Q^{1\rightarrow 2}(\epsilon) \frac{8\pi}{m_e^2} \epsilon \, d\epsilon. \tag{2.3}$$

In the inverse process, collisional de-excitation of the first excited level results from *superelastic collisions* of the type $e + A_2 \rightarrow e + A_1$. The rate per unit volume of such collisions is

$$n_e n_2 \, {}^2S_1,$$

where

$${}^2S_1 = \int_0^{\infty} f_M(\epsilon) Q^{2\rightarrow 1}(\epsilon) \frac{8\pi}{m_e^2} \epsilon \, d\epsilon. \tag{2.4}$$

In collisions of this type, since $\epsilon_2 > \epsilon_1$, the lower limit on the integration is zero. The energy difference ϵ_{12} is imparted to the free electron, hence the designation superelastic.

In addition to the foregoing, radiative processes are important in the rate equation for $\dot{n}_2$. Thus, for example, in a uniform steady-state plasma whether or not n_2/n_1 will equal its equilibrium Boltzmann value will depend on the relative magnitudes of radiative and collisional processes. The volumetric rate of depopulation of n_2 as a result of the spontaneous emission of radiation in the transition $A_2 \rightarrow A_1 + h\nu$ is

$$n_2 A_{21}.$$

Here A_{21}, an atomic constant with dimensions of $\sec^{-1}$, is the Einstein A coefficient for the transition in question (see Chapter II, Sec. 9). (Some authors define A_{21} in terms of the radiative emission per unit solid angle and hence would write $4\pi n_2 A_{21}$ in place of $n_2 A_{21}$ for the depopulation rate.) The net rate of radiative transitions between the levels in question depends on the rates of absorption and induced emission as well as on the rate of spontaneous emission. We shall discuss the role of these additional processes later. For present purposes it is sufficient to write the local *net* rate of radiative transitions as

$$n_2 A_{21}\beta_{21}.$$

The radiative-escape factor β_{21}, which is defined more precisely by equation (2.19) to follow, will be zero if locally the rate of upward $1 \rightarrow 2$ radiative transitions just counterbalances the rate of downward $2 \rightarrow 1$ radiative transitions and will be unity if the rates of absorption and induced emission can be neglected in comparison with the rate of spontaneous emission.

Combining the foregoing expressions, we obtain the simplified rate equation for $\dot{n}_2$ in the form

$$\dot{n}_2 \simeq n_e n_1 {}_1S^2 - n_2(n_e {}^2S_1 + A_{21}\beta_{21}). \tag{2.5}$$

This equation can be simplified somewhat by considerations of detailed balancing (see Chapter II, Sec. 11). In general, detailed balancing applies only to the relationship between cross sections, such as $Q^{1\rightarrow 2}$ and $Q^{2\rightarrow 1}$, which are atomic constants. If, however, the distribution function is Maxwellian, the quantities ${}_1S^2$ and ${}^2S_1$ are also related by the requirements of detailed balancing. At equilibrium, the rate $n_e n_1 {}_1S^2$ of collisional excitation of the first excited level in upward, inelastic $1 \rightarrow 2$ transitions is just balanced by the rate $n_e n_2 {}^2S_1$ of collisional de-excitation in downward,

superlastic $2 \rightarrow 1$ transitions. Accordingly, we have [see Exercise (II 11.6)]

$$\frac{{}_1S^2}{{}^2S_1} = \left(\frac{n_2}{n_1}\right)^* = \frac{g_2}{g_1} e^{-\epsilon_{12}/kT_e}, \tag{2.6}$$

where $(n_2/n_1)^*$ is the ratio of densities at equilibrium and g_2 and g_1 are the degeneracies of the levels. The use of T_e on the right-hand side of equation (2.6) is based on the following consideration: For excitation and de-excitation as a result of collisions with electrons having a Maxwellian distribution function, ${}_1S^2/{}^2S_1$ will be a function only of atomic constants and the electron temperature T_e. This function can be evaluated at equilibrium, where $T_e = T$ and where for a balance between corresponding upward and downward transitions equation (2.6) must hold.

With the use of equation (2.6) the rate equation (2.5) becomes

$$\dot{n}_2 \simeq n_e n_1 {}_1S^2 - n_2 \left[\frac{n_e {}_1S^2}{(n_2/n_1)^*} + A_{21}\beta_{21}\right] \tag{2.7a}$$

or

$$\dot{n}_2 \simeq n_e n_1 {}_1S^2 \left[1 - \frac{n_2/n_1}{(n_2/n_1)^*}\right] - n_2 A_{21}\beta_{21}. \tag{2.7b}$$

Under conditions of *complete* equilibrium, where $n_2/n_1 = (n_2/n_1)^*$ and by detailed balancing $\beta_{21} = 0$, $\dot{n}_2$ is zero. For nonequilibrium situations it can be seen from equations (2.5) and (2.7) that the relative importance of radiative and collisional rates of depopulation of the first excited level is given by the ratio

$$\frac{A_{21}\beta_{21}}{n_e {}^2S_1} = \frac{A_{21}\beta_{21}(n_2/n_1)^*}{n_e {}_1S^2}. \tag{2.8}$$

If this ratio is small, radiative processes can be neglected.

The extension of the previous considerations to the general rate equations for the $\dot{n}_k$ and $\dot{n}_e$ is a matter of straightforward bookkeeping. In the general situation, however, we must also account for the rates of ionization and recombination from and to the energy level in question. The volumetric rates of collisional ionization and three-body recombination of the form $e + A_k \rightleftharpoons e + e + A^+$ are given by

$$n_e n_k {}_kS^\lambda$$

and

$$n_e^2 n_i {}^\lambda S_k = n_e^3 {}^\lambda S_k,$$

where the symbol λ denotes the so-called continuum of states associated with the free electrons. In terms of the ionization cross section $Q^{k\to\lambda}$, we have

$$ {}_kS^{\lambda} = \int_{\epsilon_{k\lambda}}^{\infty} f_M(\epsilon) Q^{k\to\lambda} \frac{8\pi}{m_e^2} \epsilon \, d\epsilon. \tag{2.9} $$

Here $\epsilon_{k\lambda}$ is the ionization potential from the kth level. The corresponding expression for ${}^{\lambda}S_k$ in terms of the cross section $q^{\lambda\to k}$ for three-body recombination is [see Exercise (II 11.4)]

$$ {}^{\lambda}S_k = \int_0^{\infty} \int_0^{\infty} f_M(\epsilon) f_M(\epsilon') q^{\lambda\to k}(\epsilon, \epsilon') \frac{32\sqrt{2}\pi^2}{m_e^{7/2}} \epsilon\sqrt{\epsilon'} \, d\epsilon' \, d\epsilon, \tag{2.10} $$

where ϵ' denotes the energy of the second electron in the collision. A peculiarity of this process is that the cross section $q^{\lambda\to k}$, as used here, has dimensions of cm^5 rather than of cm^2. For a Maxwellian distribution, however, the use of equation (2.10) is not necessary, since by detailed balancing we have directly the relation [see Exercise (II 11.6)]

$$ \frac{{}_kS^{\lambda}}{{}^{\lambda}S_k} = \left(\frac{n_e^2}{n_k}\right)^* = \left(\frac{2\pi m_e kT_e}{h^2}\right)^{3/2} \frac{2g_i}{g_k} e^{-\epsilon_{k\lambda}/kT_e}. \tag{2.11} $$

Here g_i is the degeneracy of the ground level of the ion and the Saha equation (II 10.5a) for the kth atomic level has been used to write $(n_e^2/n_k)^*$ in terms of T_e.

We can now write the collisional contribution to $\dot{n}_k$ as

$$ \dot{n}_{k_{\text{coll}}} = n_e \left[\sum_{j<k} n_j \, {}_jS^k + \sum_{l>k} n_l \, {}^lS_k + n_e^2 \, {}^{\lambda}S_k - n_k \left(\sum_{j<k} {}^kS_j + \sum_{l>k} {}_kS^l + {}_kS^{\lambda} \right) \right] \tag{2.12a} $$

or, alternatively, as

$$ \dot{n}_{k_{\text{coll}}} = n_e \left\{ \sum_{j<k} n_j \, {}_jS^k \left[1 - \frac{n_k/n_j}{(n_k/n_j)^*} \right] + n_k \sum_{l>k} {}_kS^l \left[\frac{n_l/n_k}{(n_l/n_k)^*} - 1 \right] + n_k \, {}_kS^{\lambda} \left[\frac{n_e^2/n_k}{(n_e^2/n_k)^*} - 1 \right] \right\}. \tag{2.12b} $$

In equation (2.12a) the first three terms in the brackets account for the collisional excitation to the kth level as a result of inelastic transitions from lower levels, superelastic transitions from higher levels, and three-body recombination, respectively. The last three terms account for the collisional de-excitation from the kth level as a result of the inverse processes—superelastic transitions to lower levels, inelastic transitions to higher levels, and collisional ionization, all from the kth level. Equation

(2.12b) incorporates the detailed balancing relationships between the rate integrals and exhibits the fact that $\dot{n}_{k_{\text{coll}}} = 0$ when the ratios of the number densities have equilibrium values.

The rate integrals ${}_jS^k$ and ${}^kS_j$ in equations (2.12) are given in terms of the relevant cross sections by the following equations, which correspond to equations (2.3) and (2.4) for $1 \rightleftarrows 2$ transitions:

$$ {}_jS^k = \int_{\epsilon_{jk}}^{\infty} f_M(\epsilon) Q^{j\to k}(\epsilon) \frac{8\pi}{m_e^2} \epsilon \, d\epsilon \tag{2.13} $$

and

$$ {}^kS_j = \int_0^{\infty} f_M(\epsilon) Q^{k\to j} \frac{8\pi}{m_e^2} \epsilon \, d\epsilon. \tag{2.14} $$

The detailed-balancing relation between ${}_jS^k$ and ${}^kS_j$, which as before, holds only for $f^0 = f_M$, is [cf. equation (2.6)]

$$ \frac{{}_jS^k}{{}^kS_j} = \left(\frac{n_k}{n_j}\right)^* = \frac{g_k}{g_j} e^{-\epsilon_{jk}/kT_e}. \tag{2.15} $$

Equations (2.13) through (2.15) also apply to the rate integrals ${}_kS^l$ and ${}^lS_k$ with the appropriate change of subscripts and superscripts.

To complete the rate equation for $\dot{n}_k$ we must account, as in equation (2.5), for radiative processes. The rate per unit volume of depopulation of the kth level as a result of the spontaneous emission of radiation in the transition $\mathrm{A}_k \to \mathrm{A}_j + \mathrm{h}\nu$ is

$$ n_k A_{kj}, \tag{2.16} $$

where A_{kj} is the Einstein A coefficient for this transition. The corresponding volumetric rate of population of the kth level as a result of the absorption of radiation in the inverse transition $A_j + h\nu \to A_k$ is given by

$$ n_j B_{jk} \int_\nu \phi_{jk}(\nu) \int_0^{4\pi} I_\nu \, d\Omega \, d\nu. \tag{2.17} $$

Here B_{jk} is the Einstein B coefficient, discussed in Chapter II, Secs. 9 and 11, which is related to A_{kj} by $A_{kj}/4\pi B_{jk} = 2g_j\epsilon_{jk}^3/g_k h^2 c^2$ (h is Planck's constant and c is the speed of light). The quantity $\int_0^{4\pi} I_\nu \, d\Omega$ is the specific intensity of radiation, integrated over all solid angles, and is proportional to the local radiant energy density at the frequency ν. The line-shape factor $\phi_{jk}(\nu)$ accounts for the fact that as a result of line broadening the frequency of the radiation associated with the transition in question is not precisely fixed but varies over a narrow range. The factor $\phi_{jk}(\nu)$ has the property that $\int \phi_{jk}(\nu) \, d\nu = 1$, where the integration is extended over all frequencies of the particular line. [Strictly speaking, this normalization of ϕ_{jk}

applies to the extent that the equilibrium value of I_ν, as given by the Planck function B_ν, is approximately independent of ν over the effective range of the line. More precisely, to insure detailed balancing at equilibrium, we require that $\int \phi_{jk}(\nu) B_\nu(\nu)\, d\nu = B_\nu(h\nu = \epsilon_{jk})$, where ϵ_{jk} is fixed. See Exercise 2.4.]

In addition to transitions associated with absorption and spontaneous emission, we must in general account for the phenomenon of induced emission in which downward, $k \to j$ transitions are induced, or stimulated, by the presence of the radiation field. The rate of such transitions is proportional to n_k and to the specific intensity I_ν. If we assume that the line-shape factor for induced emission is the same as that for absorption, this rate is given by the expression (2.17) with n_j replaced by $n_k g_j/g_k$. The combined effects of absorption and induced emission can then be written as the net volumetric transition rate

$$\left(n_j - \frac{g_j}{g_k} n_k\right) B_{jk} \int_\nu \phi_{jk}(\nu) \int_0^{4\pi} I_\nu \, d\Omega \, d\nu. \tag{2.18}$$

We note in passing that if there exists a population inversion in a nonequilibrium situation such that $n_k g_j/g_k > n_j$, the specific intensity is augmented by the excess of induced emission over absorption and the radiation field gains energy at the expense of the plasma. Under suitable conditions this effect can lead to laser action. (In this case a more appropriate form of the expression (2.18) is obtained by removing the two integrations.)

Since the specific intensity I_ν depends on the emission and absorption of radiation throughout the plasma, the evaluation of expression (2.18) is coupled with considerations of radiative transfer. To avoid (or rather to make less explicit) this complexity, we introduce the local radiation-escape parameter β_{kj} defined by

$$(1 - \beta_{kj}) n_k A_{kj} \equiv \left(n_j - \frac{g_j}{g_k} n_k\right) B_{jk} \int_\nu \phi_{jk}(\nu) \int_0^{4\pi} I_\nu \, d\Omega \, d\nu. \tag{2.19}$$

In terms of β_{kj}, the net contribution to $\dot{n}_k$ as a result of the emission and absorption of radiation in $j \rightleftarrows k$ transitions is, from expressions (2.16), (2.18), and (2.19),

$$-n_k A_{kj} \beta_{kj}. \tag{2.20}$$

Under conditions for which $\beta_{kj} \simeq 0$, as when the plasma is optically thick, there exists a local balance between emission and absorption. In particular, such a balance exists for local quasiequilibrium values of n_k/n_j and I_ν. In many laboratory plasmas radiation which corresponds to

transitions for which $j > 1$ escapes from the plasma, and in this case $\beta_{kj} \simeq 1$. Transitions for which $j = 1$ terminate in the ground level; the corresponding radiation is described as resonance radiation. Since the density of absorbing ground-level atoms is relatively large, the escape parameter β_{k1} for resonance radiation is often small. Nevertheless, because A_{k1} is typically large, the product $A_{k1}\beta_{k1}$ may not be negligible in comparison with the corresponding collisional rate $n_e {}^kS_1$, particularly at lower values of n_e. The parameters β_{k1}, as well as the other values of β_{kj} if they are not known to be approximately unity, must be calculated for a particular application from the theory of radiative transfer.

We now consider the so-called free-bound radiative transitions that are associated with the photoionization and two-body recombination reactions $h\nu + A_k \leftrightarrows e + A^+$. The rate $n_e n_i A_{\lambda k}$ per unit volume of two-body recombination reactions is given in terms of the corresponding cross section $Q^{\lambda \to k}(\epsilon)$ by (see Chapter II, Sec. 11, and Exercise II 11.5)

$$n_e n_i A_{\lambda k} = n_e n_i \int_0^\infty f_M(\epsilon) Q^{\lambda \to k}(\epsilon) \frac{8\pi}{m_e^2} \epsilon \, d\epsilon. \tag{2.21}$$

We account for the absorption of radiation in the photoionization transition $h\nu + A_k \to e + A^+$ by the use of the radiation escape parameter $\beta_{\lambda k}$, where $1 - \beta_{\lambda k}$ is defined as the local ratio of the rate of photoionization reactions to the rate of two-body recombination reactions. The net contribution of these processes to $\dot{n}_k$ is then

$$+ n_e^2 A_{\lambda k} \beta_{\lambda k}, \tag{2.22}$$

where we have replaced $n_e n_i$ by n_e^2.

Combining expressions (2.20) and (2.22) and summing over all relevant transitions, we have for the total radiative contribution to $\dot{n}_k$

$$\dot{n}_{k_{\text{rad}}} = -n_k \sum_{j<k} A_{kj}\beta_{kj} + \sum_{l>k} n_l A_{lk}\beta_{lk} + n_e^2 A_{\lambda k}\beta_{\lambda k}. \tag{2.23}$$

Here the second and third terms on the right account for downward transitions that terminate in the kth state.

We can now combine equations (2.12a) and (2.23) to form an expression for the net rate of change of n_k as a result of collisional and radiative processes. Accordingly, we have

$$\dot{n}_k = n_e \sum_{j<k} n_j {}_jS^k - n_k \left[\sum_{j<k} (n_e {}^kS_j + A_{kj}\beta_{kj}) + n_e \left(\sum_{l>k} {}_kS^l + {}_kS^\lambda \right) \right] + \sum_{l>k} n_l (n_e {}^lS_k + A_{lk}\beta_{lk}) + n_e^2 (n_e {}^\lambda S_k + A_{\lambda k}\beta_{\lambda k}). \tag{2.24}$$

Here we have chosen to group terms with respect to their dependence on n_j, n_k, n_l, and n_e^2. We note that the coefficients ${}^kS_j$, ${}^lS_k$, and ${}^\lambda S_k$ can be

expressed in terms of ${}_jS^k$, ${}_kS^l$, and ${}_kS^\lambda$ by the use of equations (2.11) and (2.15).

A similar analysis leads to the corresponding expression for $\dot{n}_e$. In fact, we have already considered all the processes that contribute to $\dot{n}_e$, since these processes involve transitions to and from the bound levels. All that remains in the evaluation of $\dot{n}_e$ is a matter of bookkeeping. The result is

$$\dot{n}_e = \sum_k [n_e n_k {}_kS^\lambda - n_e^2(n_e {}^\lambda S_k + A_{\lambda k}\beta_{\lambda k})]. \tag{2.25}$$

In accord with the observation that the processes that contribute to $\dot{n}_e$ have been considered in the evaluation of $\dot{n}_k$, it can be verified from equations (2.24) and (2.25) that

$$\dot{n}_e + \sum_k \dot{n}_k = 0. \tag{2.26}$$

This reflects the physical fact that for the situation that we have considered where $\dot{n}_e = \dot{n}_i$ the total number of heavy particles, atoms plus ions, of the species in question is not changed by the nonelastic collisional and radiative processes. As a result of equation (2.26), one of the set of rate equations for $\dot{n}_k$ and $\dot{n}_e$ must be regarded as redundant.

Having obtained expressions for $\dot{n}_k$ and $\dot{n}_e$, we next turn our attention to the application of these expressions to the study of the effects of radiation escape on departures of the populations of the various excited levels and of the degree of ionization from their equilibrium values.

Exercise 2.1. Find from equations (2.7) an expression for n_2/n_1 in a uniform steady-state plasma for which, by equation (2.1), $\dot{n}_2 = 0$.

Exercise 2.2. Consider a transient situation in the absence of spatial gradients for which n_e, n_1, and T_e are constant. Initially, at time $t = 0$, n_2 is elevated above its equilibrium value. With the use of the approximate form (2.7) for $\dot{n}_2$, find an expression for n_2 as a function of time. What is a characteristic relaxation time for this process? How would your results differ if n_2 were initially depressed below its equilibrium value?

Exercise 2.3. Write the "chemical" equations, such as $e + A_j \rightarrow e + A_k$, for each type of transition that enters the rate equations (2.12a), (2.24), and (2.25).

Exercise 2.4. Verify that $\beta_{kj} = 0$ at equilibrium, when $n_k/n_j = (n_k/n_j)^*$ and the specific intensity of radiation I_ν is given by the Planck function

$$B_\nu = \frac{2h\nu^3/c^2}{e^{h\nu/kT} - 1}.$$

Exercise 2.5. Find expressions for a set of ratios, similar to those given by equation (2.8), such that if these ratios are each small, radiative processes can be neglected in the rate equations (2.24) and (2.25).

3. DEPARTURES FROM THE SAHA EQUATION

In this section we consider the equations for the level populations n_k and electron density n_e for a uniform, steady-state plasma. At first it might seem that under such conditions n_k and n_e would always be given by their equilibrium values. As a result of radiation escape, however, there may exist a nonequilibrium situation such that the ratios n_k/n_1 are not equal to $(n_k/n_1)^*$ as given by equation (2.15) and the ratio n_e^2/n_1 is not equal to the value $(n_e^2/n_1)^*$ given by the Saha equation (2.11).

Before considering the general rate equations as applied to uniform, steady-state plasmas, we shall discuss the consequences of a simple approximate formulation, the two-level model. In this model, the atoms of the ionizable species are represented in terms of only two levels, the ground level and a single excited level. From equations (2.24) and (2.25) the pertinent rate equations, those for $\dot{n}_2$ and $\dot{n}_e$, are then

$$\dot{n}_2 = 0 = n_e n_{1\,1} S^2 - n_2(n_e\,{}^2S_1 + A_{21}\beta_{21} + n_{e\,2}S^{\lambda}) + n_e^2(n_e\,{}^{\lambda}S_2 + A_{\lambda 2}\beta_{\lambda 2}) \tag{3.1}$$

and

$$\dot{n}_e = 0 = n_e(n_{1\,1}S^{\lambda} + n_{2\,2}S^{\lambda}) - n_e^2(n_e\,{}^{\lambda}S_1 + A_{\lambda 1}\beta_{\lambda 1} + n_e\,{}^{\lambda}S_2 + A_{\lambda 2}\beta_{\lambda 2}). \tag{3.2}$$

Here $\dot{n}_2$ and $\dot{n}_e$ are set equal to zero since the left-hand sides of the continuity equations for n_2 and n_e are zero for a uniform, steady-state plasma. Although the two-level model should not be regarded as an accurate representation, it can be used to demonstrate the principal qualitative effects that we wish to discuss in this section. The excited level in this model can be taken as either the actual first excited level of the ionizable species in question or as a "lumped" level of higher degeneracy selected so as to account approximately for the effects of several lower-lying levels.

It is convenient as a further approximation to neglect in the rate equation (3.1) for $\dot{n}_2$ all transitions between the excited level and the continuum. This simplification, which leads to the rate equations (2.5) and (2.7) of the previous section, will be valid if the rates of population and, separately, depopulation of n_2 in transitions from and to the continuum are small in comparison with the rates of transitions from and to the ground

level. Examination of equation (3.1) shows that sufficient conditions for this to be true are

$$ {}_2S^{\lambda} \ll {}^2S_1 \tag{3.3a} $$

and

$$ n_e^2\, {}^{\lambda}S_2 + n_e A_{\lambda 2}\beta_{\lambda 2} \ll n_1\, {}_1S^2. \tag{3.3b} $$

From physical considerations it would be expected that the condition (3.3a) would usually be satisfied, since when the ionization energy $\epsilon_{2\lambda}$ is large when compared with the electron translational energy $(3/2)kT_e$ a collision between an electron and an excited atom is more likely to result in a downward superelastic transition $(2 \rightarrow 1)$ than an ionizing transition $(2 \rightarrow \lambda)$. This is borne out by the values of ${}_2S^{\lambda}/{}^2S_1$ calculated from the quantities given in Table 1 later in this section (see Exercise 3.2). When the condition (3.3a) is satisfied, the first term on the left-hand side of the inequality (3.3b) will be much smaller than the right-hand side since, by equations (2.6) and (2.11),

$$ \frac{{}^{\lambda}S_2}{{}_1S^2} = \left(\frac{n_1}{n_e^2}\right)^* \frac{{}_2S^{\lambda}}{{}^2S_1} $$

and since for the present situation

$$ \frac{n_e^2}{n_1} \leq \left(\frac{n_e^2}{n_1}\right)^*, $$

as will be shown later in this section. Furthermore, the second term on the left-hand side of the inequality (3.3b) will also be much smaller than the right-hand side if, as is often the case, the rate of two-body recombination to n_2 is less than the rate of three-body recombination to n_2 for the *equilibrium* value of n_e (see Exercises 3.1 and 3.2).

Under these conditions the rate equation (3.1) for $\dot{n}_2$ can be replaced by [cf. equation (2.5)]

$$ \dot{n}_2 = 0 = n_e n_1\, {}_1S^2 - n_2(n_e\, {}^2S_1 + A_{21}\beta_{21}). \tag{3.4} $$

With the use of the detailed-balancing relation (2.6) the solution to equation (3.4) can then be written in the form

$$ \frac{n_2/n_1}{(n_2/n_1)^*} = \frac{1}{1 + R_2} \tag{3.5} $$

where the quantity R_2 is defined by

$$ R_2 = \frac{A_{21}\beta_{21}}{n_e\, {}^2S_1}. \tag{3.6} $$

It can be seen from equation (3.5) that the value of the quantity R_2 determines the extent to which radiation escape causes n_2/n_1 to be depressed below its equilibrium value. For very small values of β_{21} and/or large values of n_e, R_2 will be small and n_2/n_1 will approach $(n_2/n_1)^*$. In the other limit of large R_2, which in practice would occur only for relatively small values of n_e, n_2/n_1 approaches the value

$$\frac{n_2}{n_1} = \frac{n_e(n_2/n_1)^* {}^2S_1}{A_{21}\beta_{21}} = \frac{n_e {}_1S^2}{A_{21}\beta_{21}} \qquad \text{(corona limit).} \tag{3.7}$$

This situation, called the *corona limit*, is such that the level in question is excited by collisional transitions from the ground level and de-excited by radiative decay. Thus, in the corona limit radiative de-excitation is dominant in comparison with collisional de-excitation, which is just the opposite of the situation in the equilibrium limit.

The result (3.5), in addition to its role in the analysis of departures from the Saha equation which we shall discuss next, is also directly useful in the interpretation of certain spectroscopic measurements which depend on n_2. For example, spectroscopic "reversal" of the sodium D-line (5890 Å) [described by Griem (1964), sec. 13.5] is used to measure in sodium the *population temperature* T_{p_2}, which is defined in terms of n_2/n_1 by

$$\frac{n_2}{n_1} \equiv \frac{g_2}{g_1} e^{-\epsilon_{12}/kT_{p_2}}. \tag{3.8}$$

That is, T_{p_2} is a temperature that when used in the Boltzmann relation (3.8) would yield the actual value of n_2/n_1. From equations (3.5) and (2.6) it follows that the population temperature and the electron temperature are related by

$$\frac{\epsilon_{12}}{k}\left(\frac{1}{T_{p_2}} - \frac{1}{T_e}\right) = \ln(1 + R_2). \tag{3.9}$$

We note that this result (3.9) and equation (3.5) for n_2/n_1 apply whether or not the species in question, sodium in this case, is the principal source of electrons in the plasma. In practice a small amount of sodium is often added to an already partially ionized plasma for purposes of the reversal measurement.

For sodium, from Table 1 which follows, the quantities that enter equation (3.9) are: $\epsilon_{12} = 2.1$ eV, $A_{21} = 4.14 \times 10^7$ sec^{-1}, and, for example at $T_e = 2500°$K, ${}^2S_1 = 7.91 \times 10^{-8}$ cm^3/sec. To find the difference between T_{p_2} and T_e under these conditions from equation (3.9) it is necessary also to know the value of the radiation-escape parameter β_{21}, which by equation (3.6) enters in the evaluation of R_2. For a plane slab of uniform plasma of

thickness L, the solution of the equation of radiative transfer by Holstein (1951) for resonance radiation yields the result that

$$\beta_{21} = \frac{1}{3}\left[\frac{1}{\sqrt{\pi\alpha_{12}^0 x}} + \frac{1}{\sqrt{\pi\alpha_{12}^0 (L-x)}}\right] \tag{3.10}$$

where α_{12}^0 is the absorption coefficient at the center of the resonance line and where x is the distance of the point in question from the boundary of the plasma. [The absorption coefficient α_{12} is given in terms of the notation of Sec. 2 by $\alpha_{12} = n_1 B_{12} h\nu\phi_{12}(\nu)$; since typically $g_1 n_2 / g_2 n_1 \ll 1$, the effect of induced emission in equation (2.18) can be neglected. The expression (3.10) applies when the line-shape factor $\phi_{12}(\nu)$ is assumed to be determined by collision broadening.] For a laboratory plasma of thickness $L = 10$ cm at a temperature of 2500°K and with a sodium concentration n_{Na}/n of 10^{-3}, the value of β_{21} calculated from equation (3.10) at $x = L/2$ is of the order of 10^{-3}. This value is based on broadening of the line by argon-sodium collisions, but is not particularly sensitive to the perturbing species (argon). For a given relative concentration of sodium it is independent of pressure, as long as collision broadening is dominant. On the basis of equation (3.9) the departure of the population temperature T_{p_2} from T_e is shown in Fig. 1. For purposes of comparison, curves for

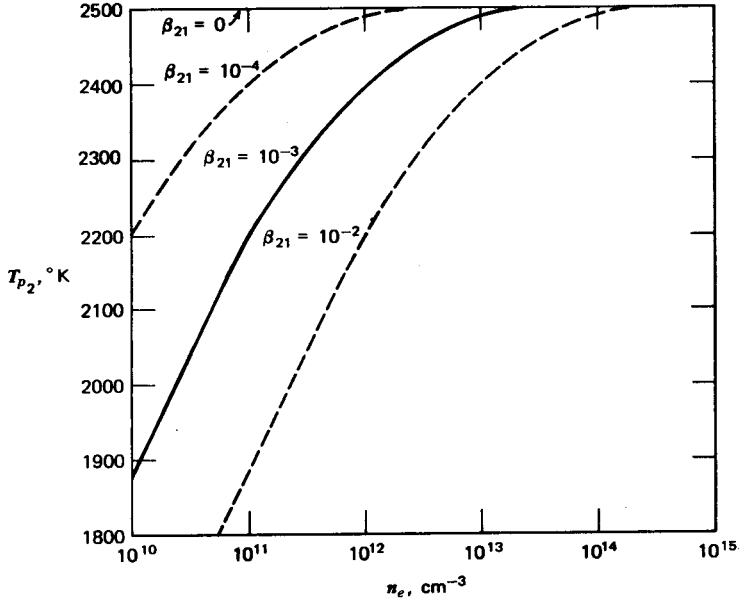


Figure 1. The effect of radiation escape on the sodium population temperature T_{p_2} at $T_e = 2500$°K, as a function of electron density for several values of the radiation-escape parameter β_{21}.

several values of β_{21} are given in the figure. It can be seen that for $\beta_{21} = 10^{-3}$ at electron densities below about $n_e = 10^{13}/\text{cm}^3$ the population temperature is no longer an accurate measure of the electron temperature. Thus even for the small values of β_{21} that result from the large absorption coefficient for resonance radiation, radiation escape can cause a significant departure of n_2/n_1 from its equilibrium value.

We now return to the coupled rate equations for the two-level model and investigate the effect of radiation escape on the departure of the electron density from its equilibrium, Saha value. The solution to the rate equation (3.2) for $\dot{n}_e$ is

$$\frac{n_e^2}{n_1}(1 + R_\lambda) = \frac{{}_1S^\lambda + ({}_2S^\lambda)n_2/n_1}{{}^\lambda S_1 + {}^\lambda S_2}. \tag{3.11}$$

The quantity R_λ which is defined by

$$R_\lambda \equiv \frac{A_{\lambda 1}\beta_{\lambda 1} + A_{\lambda 2}\beta_{\lambda 2}}{n_e({}^\lambda S_1 + {}^\lambda S_2)} \tag{3.12}$$

is a measure of the relative importance of radiative and collisional processes in the depopulation of n_e. With the use of the solution (3.5) of the rate equation for $\dot{n}_2$ and the detailed-balancing relations, the solution (3.11) for n_e^2/n_1 can be written in the form

$$\frac{n_e^2/n_1}{(n_e^2/n_1)^*} = \frac{1}{1 + R_\lambda} \times \frac{1 + ({}^\lambda S_2/{}^\lambda S_1)(n_2/n_1)/(n_2/n_1)^*}{1 + ({}^\lambda S_2/{}^\lambda S_1)}, \tag{3.13}$$

where

$$\frac{n_2/n_1}{(n_2/n_1)^*} = \frac{1}{1 + R_2}.$$

From equation (3.13) it can be seen that, according to the two-level model, the depression of n_e^2/n_1 below its equilibrium value depends on two factors, one, $(1 + R_\lambda)^{-1}$, which reflects the importance of the excess of two-body recombination over photoionization, and the other, which depends through R_2 on the escape of resonance radiation and also on the ratio ${}^\lambda S_2/{}^\lambda S_1$. Values of the various quantities that enter the evaluation of equation (3.13) are given in Table 1 for several species of interest. These values given are based on the actual first excited level for the species in question. Since most laboratory plasmas are optically thin for recombination radiation, $\beta_{\lambda 1}$ and $\beta_{\lambda 2}$ are usually nearly unity so that the quantity R_λ^0 in the last line of the table can be used in place of R_λ in equation (3.13). It can

Table 1 Approximate Values of the Quantities that Appear in the Rate Equations of the Two-Level Model for Several Atomic Species

	Cs	K	Na	A	H	
ϵ_{12}(eV)	1.432	1.615	2.104	11.650	10.196	
$\epsilon_{2\lambda}$(eV)	2.462	2.725	3.031	4.105	3.399	
g_1(—)	2	2	2	1	2	
g_2(—)	6	6	6	12	8	
g_i(—)	1	1	1	6	1	
$A_{21}(\text{sec}^{-1})$	5.07×10^7	5.72×10^7 at $T_e = 2500°K$	4.14×10^7	2.50×10^8	4.70×10^8 at $T_e = 10{,}000°K$	Approximate variation with T_e
$A_{\lambda 1}(\text{cm}^3/\text{sec})$	5.65×10^{-15}	6.60×10^{-15}	8.52×10^{-15}	1.04×10^{-15}	1.53×10^{-13}	$\propto (kT_e)^{-1/2}$
$A_{\lambda 2}(\text{cm}^3/\text{sec})$	8.53×10^{-15}	9.94×10^{-15}	1.16×10^{-14}	3.04×10^{-15}	7.62×10^{-14}	$(kT_e)^{-1/2}$
$^2S_1(\text{cm}^3/\text{sec})$	7.20×10^{-7}	2.42×10^{-7}	7.91×10^{-8}	3.09×10^{-9}	8.91×10^{-9}	$(kT_e)^{1/2}$
$A_{21}/{}^2S_1 = n_e R_2/\beta_{21}$ (cm^{-3})	7.05×10^{13}	2.36×10^{14}	5.22×10^{14}	8.10×10^{16}	5.27×10^{16}	$(kT_e)^{-1/2}$
$(n_e^2/n_1)^*(\text{cm}^3)$	4.17×10^{12}	5.10×10^{11}	1.33×10^{10}	3.24×10^{14}	3.50×10^{14}	$\exp(-\epsilon_{1\lambda}/kT_e)$
$^\lambda S_1(\text{cm}^6/\text{sec})$	1.48×10^{-27}	1.46×10^{-27}	1.45×10^{-27}	3.35×10^{-29}	1.77×10^{-30}	$(kT_e)^{-1}$
$^\lambda S_2(\text{cm}^6/\text{sec})$	4.70×10^{-27}	4.61×10^{-27}	4.56×10^{-27}	4.91×10^{-28}	1.75×10^{-28}	$(kT_e)^{-1}$
$^\lambda S_2/{}^\lambda S_1$(—)	3.18	3.16	3.15	14.68	98.8	—
$\dfrac{A_{\lambda 1} + A_{\lambda 2}}{{}^\lambda S_1 + {}^\lambda S_2} \equiv n_e R_\lambda^0(\text{cm}^{-3})$	2.29×10^{12}	2.72×10^{12}	3.34×10^{12}	7.75×10^{12}	1.29×10^{15}	$(kT_e)^{1/2}$

be seen that for the conditions given in the table ${}^{\lambda}S_2/{}^{\lambda}S_1$ is greater than unity. Hence, in equation (3.13) we have roughly

$$\frac{1 + ({}^{\lambda}S_2/{}^{\lambda}S_1)(n_2/n_1)/(n_2/n_1)^*}{1 + ({}^{\lambda}S_2/{}^{\lambda}S_1)} \simeq \frac{n_2/n_1}{(n_2/n_1)^*} + \frac{{}^{\lambda}S_1}{{}^{\lambda}S_2}.$$

Thus n_e^2/n_1 will be approximately proportional to n_2/n_1 and hence sensitive to the escape of resonance radiation to the same degree as n_2/n_1 to the extent that ${}^{\lambda}S_1/{}^{\lambda}S_2 < (n_2/n_1)/(n_2/n_1)^*$.

From values such as those given in Table 1, equation (3.13) can be used to find the departure of the degree of ionization, expressed as n_e^2/n_1, from its equilibrium value as a function of electron density and the radiation-escape parameters β_{21}, $\beta_{\lambda 1}$, and $\beta_{\lambda 2}$. Curves of this nature are shown in Fig. 2 for the conditions of Table 1 and for $\beta_{21} = 10^{-3}$ and $\beta_{\lambda 1} = \beta_{\lambda 2} = 1$. As we noted earlier, these results based on the two-level model, although indicative of the actual qualitative behavior, should not be taken as numerically accurate.

As in the case of n_2/n_1 [see equation (3.5) and (3.7)] two limiting situations are apparent for equation (3.13). As before, in the equilibrium limit of large n_e, collisional processes are dominant in comparison with radiative processes and $n_e^2/n_1 = (n_e^2/n_1)^*$. On the other hand, in the corona limit for relatively small n_e depopulation of both n_2 and n_e occurs primarily as a result of radiative processes. In this limit for which R_λ and R_2 are large, equation (3.13) reduces to

$$n_e^2/n_1 = \frac{(n_e^2/n_1)^* n_e {}^{\lambda}S_1}{A_{\lambda 1}\beta_{\lambda 1} + A_{\lambda 2}\beta_{\lambda 2}}$$

$$= \frac{n_{e\,1} S^{\lambda}}{A_{\lambda 1}\beta_{\lambda 1} + A_{\lambda 2}\beta_{\lambda 2}} \qquad \text{(corona limit).} \qquad (3.14)$$

Similarly to the corresponding equation (3.7) for n_2/n_1 in the corona limit, equation (3.14) expresses a balance between collisional ionization from the ground level and radiative decay through two-body recombination. This situation is reflected in Fig. 2 by the straight-line portions of the curves at the lower number densities.

The use of equation (3.13) and Fig. 2 is straightforward when n_e, T_e, and the radiation-escape parameters, β_{12}, etc., are specified and the departure of the degree of ionization from its equilibrium value is to be found. If, on the other hand, the ground-state number density n_1 is known and the electron density is required, the solution involves a cubic equation for n_e, as can be seen most readily from the original rate equations (3.4) and (3.2). The solution in this case can be obtained in several ways, for example, by

iteration. The results of such a calculation for the conditions corresponding to those of Fig. 2 are shown in Fig. 3. With the results of Fig. 2 available, the calculations necessary for Fig. 3 are easily performed. The qualitative effects given by equation (3.13) and shown in Fig. 2 are reflected in Fig. 3 as well. Now, however, the equilibrium limit of a dense plasma appears for large values of the ground-state number density n_1 and the corona limit of a tenuous plasma appears for small values of n_1.

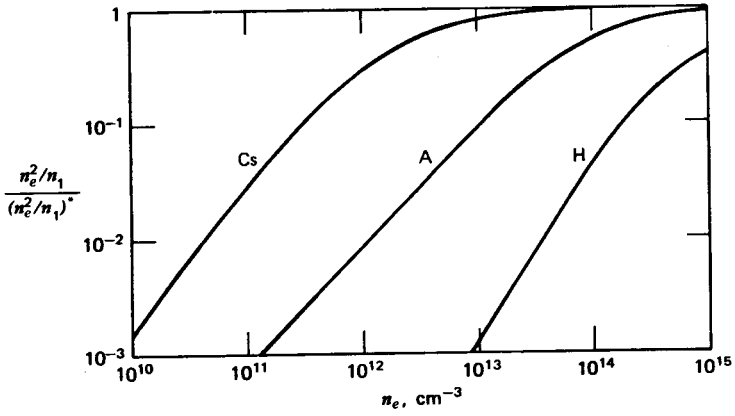


Figure 2. Departure of n_e^2/n_1 from its equilibrium value according to the two-level model for Cs at $T_e = 2500°K$ and A and H at 10,000°K based on the values of Table 1, $\beta_{21} = 10^{-3}$, and $\beta_{\lambda 1} = \beta_{\lambda 2} = 1$.

The foregoing discussion has been based on the two-level model, which illustrates the situation of interest in a relatively simple way. For accurate calculations of the effect of radiation escape on departures from the Saha equation, however, it is necessary to consider more complete rate equations. For a uniform, steady-state plasma the continuity equations (2.1) and (2.2) for n_k and n_e, with equations (2.24) and (2.25) for $\dot{n}_k$ and $\dot{n}_e$, are

$$\dot{n}_k = 0 = n_e \sum_{j<k} n_j {}_j S^k - n_k \left[\sum_{j<k} (n_e {}^k S_j + A_{kj}\beta_{kj}) + n_e \left(\sum_{l>k} {}_k S^l + {}_k S^\lambda \right) \right] + \sum_{l>k} n_l (n_e {}^l S_k + A_{lk}\beta_{lk}) + n_e^2 (n_e {}^\lambda S_k + A_{\lambda k}\beta_{\lambda k}) \tag{3.15}$$

and

$$\dot{n}_e = 0 = \sum_k [n_e n_k {}_k S^\lambda - n_e^2 (n_e {}^\lambda S_k + A_{\lambda k}\beta_{\lambda k})], \tag{3.16}$$

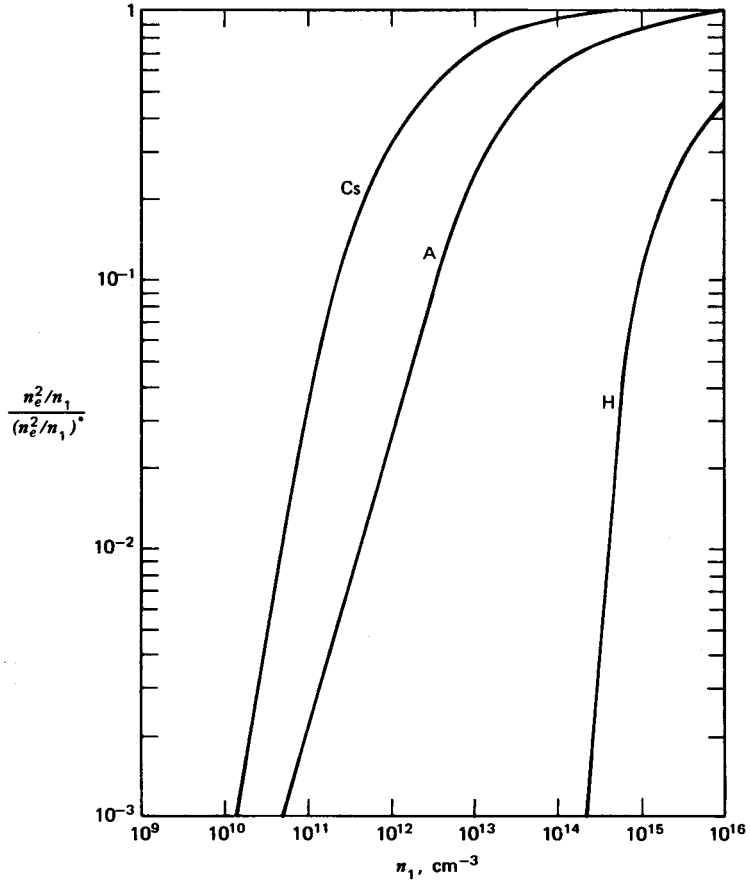


Figure 3. Departure of n_e^2/n_1 from its equilibrium value for the conditions of Fig. 2, expressed as a function of ground-state number density n_1.

where, from the detailed-balancing relations (2.15) and (2.11), we have

$$\frac{{}_jS^k}{{}^kS_j} = \left(\frac{n_k}{n_j}\right)^*, \qquad \frac{{}_kS^l}{{}^lS_k} = \left(\frac{n_l}{n_k}\right)^*,$$

and (3.17)

$$\frac{{}_kS^\lambda}{{}^\lambda S_k} = \left(\frac{n_e^2}{n_k}\right)^*.$$

As with the two-level model, the solution to equations (3.15) and (3.16) is readily obtained in two limits, which depend on the relative importance

of collisional and radiative processes. In the quasiequilibrium limit when collisional processes are dominant, it is readily verified with the aid of equations (3.17) that the solution to equations (3.15) and (3.16) is [see equation (2.12b)]

$$\left(\frac{n_k}{n_j}\right) = \left(\frac{n_k}{n_j}\right)^*$$

and (3.18)

$$\frac{n_e^2}{n_k} = \left(\frac{n_e^2}{n_k}\right)^*$$

for all values of k and j. In particular, we then have

$$\frac{n_k}{n_1} = \left(\frac{n_k}{n_1}\right)^*$$

and (3.19)

$$\frac{n_e^2}{n_1} = \left(\frac{n_e^2}{n_1}\right)^*.$$

It is easily seen that *sufficient* conditions for the validity of the quasiequilibrium solution (3.18) are, for all values of k and j,

$$\frac{A_{kj}\beta_{kj}}{n_e {}^kS_j} \ll 1$$

and (3.20)

$$\frac{A_{\lambda k}\beta_{\lambda k}}{n_e {}^{\lambda}S_k} \ll 1.$$

These relations, which will be satisfied for large electron densities and for small values of the radiation-escape parameters β_{kj} and $\beta_{\lambda k}$, correspond to the conditions $R_2 \ll 1$ and $R_\lambda \ll 1$ for the two-level model [see equations (3.5) and (3.11)]. Thus, when collisional processes are dominant over radiative processes, a quasiequilibrium situation exists for which the electron density is given by the Saha equation. We note that for the model that we have considered the Saha equation under these conditions is to be evaluated at the electron temperature T_e, whether or not T_e is equal to the heavy-particle translational temperature. This results from the fact that the collisional processes in question involve the translational energy of only the free electrons.

The second limiting situation, the corona limit, occurs when the electron density is low and the radiation-escape parameters are not small. In the limit of very low electron density, the dominant processes for depopulation of n_k and n_e are downward radiative transitions, and it is found that

n_k and n_e are determined by a balance between the rates of collisional population from the ground level and radiative decay. Thus in the corona limit we have

$$\frac{n_k}{n_1} = \frac{n_e\, {}_1S^k}{\sum_{j<k} A_{kj}\beta_{kj}} \tag{3.21a}$$

and (corona limit)

$$\frac{n_e}{n_1} = \frac{{}_1S^\lambda}{\sum_{k\geq 1} A_{\lambda k}\beta_{\lambda k}}. \tag{3.21b}$$

As with the corresponding equations (3.7) and (3.14) for the two-level model, the relations (3.21) are markedly different from the equilibrium relations (2.14) and (2.10), particularly in that they depend in detail on the collisional and radiative rate coefficients. Equations (3.21) for the corona limit, however, should be regarded as primarily of formal and illustrative significance. In actual situations, particularly for laboratory plasmas, at low electron densities effects not considered in our model, such as excitation as a result of atom-atom collisions and departures from a Maxwellian distribution, become important and equations (3.21) must be modified accordingly. Furthermore, for the higher-lying levels collisional rates of depopulation remain significant in comparison to radiative rates even at relatively low electron densities, so that equation (3.21a) will not apply to such levels. We shall discuss this point in greater detail later in this section.

For conditions intermediate between the equilibrium limit and the corona limit, the calculation of the level populations and the electron density is more complicated. In principle, this calculation can be carried out as follows:

For a given value of the electron temperature and known values of the collisional and radiative rate coefficients, including the radiation-escape parameters, equations (3.15) and (3.16) represent $q + 1$ equations for the $q + 1$ densities, n_k ($k = 1, 2, \ldots q$) and n_e. The manner in which the number q of levels can be specified in actual calculations will be discussed later in this section. As a result of conservation of the total number of excited atoms and ions we have $\sum_k \dot{n}_k + \dot{n}_e = 0$, so that one of these $q + 1$ equations is redundant. Correspondingly, in a given physical situation either the total number of excited atoms and ions or the number of atoms in the ground state or some other relation between the $q + 1$ densities must be specified. Since equations (3.15) and (3.16) are linear in the values of n_k but cubic in n_e, it is convenient for calculational purposes to regard n_e as specified. If, for example, we omit the equation $\dot{n}_1 = 0$, we then have q linear, inhomogeneous algebraic equations [$\dot{n}_k = 0$ ($2 \leq k \leq q$), and $\dot{n}_e = 0$]

for the q variables $n_k (1 \leq k \leq q)$, as functions of n_e and T_e. These equations can be solved by Cramer's rule and, if necessary, the actual physically specified conditions satisfied by iteration.

In practice, some approximate simplification of the calculation is almost always employed. For example, in place of a realistic value for the number q of excited states, the electronic excitation structure of the atom is approximated in some calculations by a small number of levels, as in the two-level model. These levels can be taken as corresponding to the actual lower-lying levels of the atom, so that the higher levels are ignored, or the higher levels can be combined, approximately, into one or more levels which are assigned artificially large degeneracies (see, e.g., BenDaniel and Tamor, 1962; Shaw, et al., 1966; and Dugan, Lyman and Albers, 1966). In addition, consideration of the relative magnitudes of the various collisional and radiative rate coefficients can be used to simplify equations (3.15) and (3.16) by the omission of certain terms. Thus Zel'dovich and Raizer (1966), Sec. VI 18, argue that under certain conditions, for at least the higher-lying levels, all collisional processes can be neglected except those that involve transitions between adjacent levels. We shall not pursue further the consequences of these considerations. There is, however, one particular simplification of the rate equations, with important physical consequences, which we shall now discuss in somewhat greater detail.

It is found experimentally that over a wide range of conditions the higher-lying excited atomic levels are in Saha equilibrium with the free electrons (see, e.g., Hinnov and Hirschberg, 1962). That is, even in non-equilibrium situations, the populations of excited levels that lie above a certain level, say the pth level, are related to the electron density by Saha equations of the form [cf. equation (2.11)]

$$\frac{n_e^2}{n_{k \geq p}} = \left(\frac{2\pi m_e kT_e}{h^2}\right)^{3/2} \frac{2g_i e^{-\epsilon_{ki}/kT_e}}{g_k}. \tag{3.22}$$

This results from the fact that the relative magnitude of collisional rates in comparison with radiative rates becomes greater for higher-lying levels for which the energy difference between adjacent levels becomes smaller.

To show that Saha equilibrium with the free electrons applies for higher-lying levels, we estimate the relative magnitudes of the rates of collisional and radiative transitions in establishing the populations of such levels. To the extent that collisional transitions between higher-lying levels and the continuum dominate these rates, such levels will be in collisional equilibrium among themselves and with the free electrons, and equation (3.22) will apply. To this end we use the semiempirical representation given by Allen (1963), Sec. 18, for the excitation cross section $Q^{k \to l}$, which is found to

correlate with the Einstein coefficient A_{kl}, to estimate that roughly at least for higher-lying levels [see also Griem (1964), Sec. 6.7, and Chapter II, Secs. 4 and 8]

$$\frac{n_e {}_kS^l}{A_{lk}} \propto n_e \frac{e^{-\epsilon_{kl}/kT_e}}{\epsilon_{kl}^3}. \tag{3.23a}$$

Using the detailed-balancing relation (2.15), we then have

$$\frac{n_e {}^lS_k}{A_{lk}} \propto \frac{n_e}{\epsilon_{kl}^3}. \tag{3.23b}$$

Although the exact form of this result depends on the representation adopted for the excitation cross section, for any reasonable cross section the conclusion that the ratio (3.23b) increases as ϵ_{kl} decreases remains the same. Thus, depending on n_e, above a certain level for which $\epsilon_{k+1} - \epsilon_k$ is sufficiently small, collisional processes will tend to dominate radiative processes in transitions between adjacent levels. Furthermore, by equations (3.23) collisional rates are greater between neighboring levels. Therefore, if the rate of collisional population of the pth level from adjacent levels is much greater than the rate of radiative depopulation of that level, that is, if

$$\frac{n_{p+1} n_e {}^{p+1}S_p}{n_p \sum_{j<p} A_{pj}} \sim \frac{n_e {}_pS^{p+1}}{\sum_{j<p} A_{pj}} \gg 1, \tag{3.24}$$

the level in question will be in collisional equilibrium with adjacent levels. In addition, since $\sum_{j<k} A_{kj}$ becomes smaller for greater k, whereas ${}_kS^{k+1}$ becomes larger (see Griem, 1964, Sec. 6.7), all levels for which $k \geq p$ will be in collisional equilibrium. Finally, by similar considerations, such levels will be in close collisional communication with the free electrons, so that for $k \geq p$ the populations n_k will be given by the Saha equation (3.22). We note that it follows from equation (3.22) that for such levels we also have

$$\frac{n_k}{n_j} = \frac{g_k}{g_j} e^{-\epsilon_{jk}/kT_e} \qquad (k, j \geq p). \tag{3.25}$$

These conclusions are supported by the detailed calculations of Bates and Kingston (1963), which we shall discuss in the following section.

We now suppose that for a given situation it has been found that the inequality (3.24) applies for a certain level, the pth level, and all higher levels, so that the Saha equations (3.22) hold for these levels. We then ask what are the resulting modifications of the rate equations (2.24) and (2.25). Since by equation (3.22) we then have $n_k \propto n_e^2$ for $k \geq p$, we can omit the rate equations (2.24) for $k \geq p$ and can in the remaining equations

group the terms that involve these states with the terms that involve the continuum. To this end we introduce the definitions

$$
\begin{aligned}
{}_kS^{\lambda'} &\equiv {}_kS^{\lambda} + \sum_{l \geq p} {}_kS^{l}, \\
{}^{\lambda'}S_k &\equiv {}^{\lambda}S_k + \sum_{l \geq p} \left(\frac{n_l}{n_e^2}\right)^* {}^{l}S_k = \left(\frac{n_k}{n_e^2}\right)^* {}_kS^{\lambda'},
\end{aligned}
\tag{3.26}
$$

and

$$
A_{\lambda' k}\beta_{\lambda' k} \equiv A_{\lambda k}\beta_{\lambda k} + \sum_{l \geq p} \left(\frac{n_l}{n_e^2}\right)^* A_{lk}\beta_{lk}.
$$

Equation (2.24) for $k < p$ then becomes, with the use of equation (3.22),

$$
\begin{aligned}
\dot{n}_k = n_e \sum_{j<k} n_j {}_jS^k - n_k \left[\sum_{j<k} (n_e {}^kS_j + A_{kj}\beta_{kj}) + n_e \left(\sum_{p>l>k} {}_kS^l + {}_kS^{\lambda'} \right) \right] \\
+ \sum_{p>l>k} n_l (n_e {}^lS_k + A_{lk}\beta_{lk}) + n_e^2 (n_e {}^{\lambda'}S_k + A_{\lambda' k}\beta_{\lambda' k}). \qquad (k < p)
\end{aligned}
\tag{3.27}
$$

With respect to equation (2.25) for $\dot{n}_e$, we might directly apply equation (3.22) and detailed balancing [equation (2.11)] to eliminate the terms that involve $n_{k \geq p}$ (see Exercise 3.5). It is preferable to write instead, from equations (2.24), (2.25), (3.22), and (3.26),

$$
\dot{n}_e + \sum_{l \geq p} \dot{n}_l = \sum_{k<p} [n_e n_k {}_kS^{\lambda'} - n_e^2 (n_e {}^{\lambda'}S_k + A_{\lambda' k}\beta_{\lambda' k})] \tag{3.28a}
$$

where, by differentiation of equation (3.22),

$$
\dot{n}_e + \sum_{l \geq p} \dot{n}_l = \dot{n}_e + \sum_{l \geq p} \left(\frac{n_l}{n_e^2}\right)^* 2 n_e \dot{n}_e = \left[1 + 2 \sum_{l \geq p} \left(\frac{n_l}{n_e}\right) \right] \dot{n}_e. \tag{3.28b}
$$

Since, under many conditions

$$
\sum_{l \geq p} \frac{n_l}{n_e} \ll 1,
$$

we then have

$$
\dot{n}_e \simeq \sum_{k<p} [n_e n_k {}_kS^{\lambda'} - n_e^2 (n_e {}^{\lambda'}S_k + A_{\lambda' k}\beta_{\lambda' k})]. \tag{3.29}
$$

Equations (3.27) and (3.28) or (3.29) compromise p equations for the p densities, $n_1, n_2, \ldots, n_{p-1}$, and n_e. As before, one of these equations must be regarded as redundant and one of the densities as specified.

Physically, the foregoing considerations indicate that transitions between a lower level and the pth level or above are equivalent in effect to transitions

between the lower level and the continuum, as can be seen from a comparison of equation (3.29) and the previous equation (2.25). For a uniform, steady-state plasma the solution for $n_{k<p}$ and n_e to equations (3.27) and (3.28) or (3.29) with $\dot{n}_k = 0$ and $\dot{n}_e = 0$ can be accomplished as outlined earlier. Now, however, the condition (3.24) fixes the number of levels that must be considered, which was left imprecise in the previous discussion.

With the aid of equations (3.27) and (3.29), we discuss in the following section ionization and recombination rates in transient plasmas.

Exercise 3.1. Show that for the conditions of this section when the inequality (3.3a) is satisfied and when

$$A_{\lambda 2}\beta_{\lambda 2} \leq \left[\left(\frac{n_e^2}{n_1}\right)^* n_1\right]^{1/2} {}^{\lambda}S_2,$$

that

$$n_e A_{\lambda 2}\beta_{\lambda 2} \ll n_{1\,1}S^2,$$

so that the inequality (3.3b) is also satisfied. This in effect replaces the condition (3.3b) with the condition that for the equilibrium value of n_e the rate of two-body recombination to n_2 be less than the rate of three-body recombination.

Exercise 3.2. With the use of the quantities given in Table 1, the detailed balancing relations, and Exercise 3.1, find for the species cited the values of n_1 for which the inequalities (3.3a) and (3.3b) are satisfied for the temperatures given in the table.

Exercise 3.3. For cesium at 2500°K and hydrogen at 10,000°K, prepare plots, similar to Fig. 1, that show the effect of radiation escape on the population temperature as a function of n_e, on the basis of the two-level model.

Exercise 3.4. From Fig. 3 find the pressures of, respectively, Cs at 2500°K and A and H at 10,000°K below which (according to the two-level model) n_e will be less than one half its equilibrium value, assuming that $\beta_{21} = 10^{-3}$ and $\beta_{\lambda 1} = \beta_{\lambda 2} = 1$. (Since in general the plasma may contain other, non-ionized species, these pressures are the partial pressure of the ionizable species in question.)

Exercise 3.5. Consider the case where the pth level and all higher levels are (approximately) in equilibrium with the free electrons, so that the Saha equation (3.22) holds (approximately) for these levels. Use equations (3.22) and (2.11) to eliminate the terms that involve $n_{k>p}$ in the rate equation (2.25) for $\dot{n}_e$, and compare your result with equations (3.28) and (3.29). Can you interpret physically the differences between these results

and explain why equations (3.28) and (3.29) are to be preferred? Suggestion: Apply the two-level model [in the form of equations (2.5) and (3.2)] to this situation. What are the conditions that $n_2/n_e^2 \simeq (n_2/n_e^2)^*$?

4. IONIZATION AND RECOMBINATION RATES

When plasma conditions change rapidly, as, for example, in a time-varying stationary plasma or in the rapid expansion of a plasma through a nozzle, departures from the Saha equation can result from finite ionization and recombination rates. In this section we discuss the application of the equations for $\dot{n}_k$ and $\dot{n}_e$ to the study of such rates.

In general, ionization and recombination phenomena are described by the continuity equations (2.1) and (2.2) for n_k and n_e, with $\dot{n}_k$ and $\dot{n}_e$ given by the rate equations (2.24) and (2.25). To make the situation more manageable, we impose two simplifications on these equations. First, we assume as in Sec. 3 that the pth excited level and all higher levels are in Saha equilibrium with the free electrons, so that equation (3.22) holds for these levels. The rate equations (2.24) and (2.25) for $\dot{n}_k$ and $\dot{n}_e$ are then replaced by equations (3.27) and (3.28) or (3.29). The appropriate value for p in a given situation can be found approximately from the inequality (3.24) or *a posteriori* from the solution of the rate equations for n_k. In the latter case, a value of p is chosen, the rate equations are then solved for n_k, and the values of n_k are examined to verify that they satisfactorily approach Saha-equilibrium values as the pth level is approached from below.

The second simplification is based on the observation that the time for adjustment, or relaxation, of the excited-level populations ($k > 1$) is in many cases much more rapid than the ionization or recombination time. Accordingly, the values of n_k for $k > 1$ will tend to be in balance with the instantaneous values of n_e and n_1, and we can set

$$\dot{n}_k \simeq 0, \qquad k > 1. \tag{4.1}$$

This approximation can be made plausible in particular situations by means of order-of-magnitude estimates, based on equations (3.27) and (3.29), of the characteristic times τ_e and τ_k for adjustment of n_e and n_k. For example, if n_e and n_k are established mainly by collisional processes and if we estimate τ_e and τ_k from the rates of population, we have

$$\tau_e \sim \left(\frac{\dot{n}_e}{n_e}\right)^{-1} \sim \frac{1}{\sum_{l<p} n_l {}_l S^{\lambda'}}$$

and

$$\tau_k \sim \left(\frac{\dot{n}_k}{n_k}\right)^{-1} \sim \frac{n_k}{n_e\left(\sum_{j<k} n_j {}_j S^k + \sum_{p>l>k} n_l {}_l S_k + n_e^2 {}^{\lambda'} S_k\right)}.$$

The ratio of the characteristic times is then

$$\frac{\tau_k}{\tau_e} \sim \frac{n_k}{n_e}\left[\frac{\sum_{l<p} n_l\, {}_lS^{\lambda'}}{\sum_{j<k} n_j\, {}_jS^k + \sum_{p>l>k} n_l\, {}^lS_k + n_e^2\, {}^{\lambda'}S_k}\right].$$

Under most conditions the factor in brackets in equation (4.2) is less than unity. That this is physically reasonable can be seen from the following argument: In both the numerator and the denominator of this ratio there are terms that are proportional to each of the densities n_l for $1 \leq l < p$ (as well as an additional term which is proportional to n_e^2 in the denominator). If the coefficient of each of these terms in the denominator is at least as large as the corresponding term in the numerator, the ratio in question will be less than unity. This will be the case if ${}_lS^k \geq {}_lS^{\lambda'}$ and ${}^lS_k \geq {}_lS^{\lambda'}$. Physically, these inequalities will usually hold, since in a collision between an electron and an excited atom with $l < p$ the probabilities that the atom will be excited to a higher level or de-excited to a lower level are usually greater than the probability that the atom will be ionized. From equation (4.2) we then find that

$$\tau_k \ll \tau_e \qquad \text{if} \qquad n_k \ll n_e, \tag{4.3}$$

as is usually the case for $k > 1$.

With these simplifications we can use equation (3.29) for $\dot{n}_e$, and the relevant equations become

$$\frac{\partial n_e}{\partial t} + \nabla \cdot [n_e(\mathbf{u} + \mathbf{U}_e)] = \dot{n}_e, \tag{4.4}$$

where

$$\dot{n}_e = \sum_{k<p} [n_e n_k\, {}_kS^{\lambda'} - n_e^2({}^{\lambda'}S_k + A_{\lambda' k}\beta_{\lambda' k})] \tag{4.5}$$

and

$$\begin{aligned}\dot{n}_k = 0 = {} & n_e \sum_{j<k} n_j\, {}_jS^k \\ & - n_k\left[\sum_{j<k}({}^kS_j + A_{kj}\beta_{kj}) + n_e\left(\sum_{p>l>k} {}_kS^l + {}_kS^{\lambda'}\right)\right] \\ & + \sum_{p>l>k} n_l({}^lS_k + A_{lk}\beta_{lk}) + n_e^2({}^{\lambda'}S_k + A_{\lambda' k}\beta_{\lambda' k}) \\ & (1 < k < p). \end{aligned} \tag{4.6}$$

As in equations (3.26), the symbol λ' denotes the set composed of the continuum and the pth and higher levels, which are assumed to be in Saha

equilibrium with the free electrons. The detailed-balancing relations between the rate coefficients are, from equations (2.15) and (3.26),

$$\frac{{}_jS^k}{{}^kS_j} = \left(\frac{n_k}{n_j}\right)^* \tag{4.7a}$$

and

$$\frac{{}_kS^{\lambda'}}{{}^{\lambda'}S_k} = \left(\frac{n_e^2}{n_k}\right)^*. \tag{4.7b}$$

As before, we regard the equation for $\dot{n}_1$ as redundant. We note, however, that it follows from $\dot{n}_k = 0$ $(k > 1)$ that $\dot{n}_1 = -\dot{n}_e$, so that an equivalent procedure would be to omit equation (4.5) for $\dot{n}_e$ and retain the equation for $\dot{n}_1$.

The general form of the solution to the foregoing equations can be seen as follows: Assuming that the coefficients $A_{kj}\beta_{kj}$, ${}_jS^k(T_e)$, and ${}_kS^{\lambda'}(T_e)$ are known, the homogeneous algebraic equations (4.6) can be solved for the various values $n_k(1 < k < p)$ as functions of n_e, n_1, and T_e. With the use of these values, equation (4.5) then yields an expression $\dot{n}_e = \dot{n}_e(n_e, n_1, T_e)$. It is this expression, when used with equation (4.4), that describes the ionization and recombination process.

As in Sec. 3, we now consider formally the expressions for $\dot{n}_e$ in the limits of high and low electron density. In the corona limit of low electron density, depopulation of the various levels and of n_e occurs primarily as a result of radiative processes rather than by collisional processes. To the extent, then, that $\dot{n}_e$ is established by a balance between collisional ionization from the ground level and two-body radiative recombination [cf. equation (3.21b)] we have directly

$$\dot{n}_e = n_e n_1 {}_1S^{\lambda'} - n_e^2 \sum_{k<p} A_{\lambda' k}\beta_{\lambda' k} \qquad \text{(corona limit)} \tag{4.8}$$

This result is independent of the solution of equation (4.6) for n_k, the values of which are given by equation (3.21a) as before. Again we must emphasize, however, that the results for the corona limit such as equation (4.8) may be inaccurate when applied to actual plasmas, since at low electron densities, additional effects such as atom-atom excitation and departures from a Maxwellian distribution for the free electrons may be important. Furthermore, irrespective of these effects, Bates and Kingston (1964) have found that very low electron densities are required in order that the recombination rate [corresponding to the negative term in equation (4.8)] approach its value for purely radiative recombination. Thus, for example, they find that for a hydrogen plasma at $T_e = 250°\text{K}$ collisional effects increase the recombination rate by more than a factor of two for values of n_e greater than about $2 \times 10^5\text{cm}^{-3}$.

In the limit of high electron density, collisional processes are dominant in comparison with radiative processes, and the recombination mechanism is one of three-body recombination. In this limit, in accord with the inequalities (3.20), the radiative terms can be omitted in the rate equations (4.5) and (4.6), which then become

$$\dot{n}_e = \sum_{k<p} (n_e n_k {}_k S^{\lambda'} - n_e^3 {}^{\lambda'}S_k) \tag{4.9}$$

and

$$\sum_{1<j<k} n_j {}_j S^k - n_k \left(\sum_{j<k} {}^k S_j + \sum_{p>l>k} {}_k S^l + {}_k S^{\lambda'} \right) + \sum_{p>l>k} n_l {}^l S_k$$
$$= -(n_1 {}_1 S^k + n_e^2 {}^{\lambda'}S_k). \qquad (k > 1) \tag{4.10}$$

The densities n_k for $1 < k < p$ can be found from the rate equations (4.10) in terms of n_e, n_1, and T_e. We have written the terms in n_1 and n_e on the right-hand side of equation (4.10) to emphasize the fact that n_1 and n_e are regarded here as parameters. With the use of Cramer's rule and the properties of determinants, it can be seen by inspection that the solution to equations (4.10) must be of the form

$$n_k = a_k' n_1 + a_k n_e^2. \tag{4.11}$$

The quantities a_k' and a_k depend on the electron temperature through the rate coefficients ${}_j S^k$ and ${}_k S^{\lambda'}$, but not the densities, and can be expressed as ratios of determinants. Furthermore, since radiative processes are omitted from the present calculations, whenever $n_e^2/n_1 = (n_e^2/n_1)^*$ we must have $n_k/n_1 = (n_k/n_1)^*$ [see equation (2.12b)]. It follows that

$$a_k' = \left(\frac{n_k}{n_1}\right)^* - a_k \left(\frac{n_e^2}{n_1}\right)^*.$$

Equation (4.11) then becomes

$$\frac{n_k}{n_1} = \left(\frac{n_k}{n_1}\right)^* + a_k \left[\frac{n_e^2}{n_1} - \left(\frac{n_e^2}{n_1}\right)^* \right]. \tag{4.12}$$

We can now use this result in equation (4.9) to obtain for $\dot{n}_e$

$$\dot{n}_e = n_e n_1 \sum_{k<p} \left[\left(\frac{n_k}{n_1}\right)^* - a_k \left(\frac{n_e^2}{n_1}\right)^* \right] {}_k S^{\lambda'}$$
$$- n_e^3 \sum_{k<p} ({}^{\lambda'}S_k - a_k {}_k S^{\lambda'}).$$

Applying equation (4.7b), we have finally

$$\dot{n}_e = \alpha\left[n_1\left(\frac{n_e^2}{n_1}\right)^* - n_e^2\right], \tag{4.13}$$

where the overall recombination coefficient α is given in terms of the quantities a_k of equation (4.12) by

$$\begin{aligned}\alpha &= n_e \sum_{k<p}\left[\left(\frac{n_k}{n_e^2}\right)^* - a_k\right]_k S^{\lambda'} \\ &= n_e \sum_{k<p} ({}^{\lambda'}S_k - a_{k\,k}S^{\lambda'}).\end{aligned} \tag{4.14}$$

It can be seen that in accord with macroscopic reasoning the expression (4.13) for $\dot{n}_e$ consists of one term which is proportional to $n_e n_1$ and another which is proportional to $-n_e^3$. Furthermore, equation (4.13) yields the steady-state solution $n_e^2/n_1 = (n_e^2/n_1)^*$ when $\dot{n}_e = 0$. This recovery of Saha equilibrium at the steady state is a consequence of the fact that radiative processes have been omitted. We note that for the present collision-dominated situation the linear dependence of α on n_e, shown by equations (4.14), could have been incorporated explicitly in equation (4.13) by the use of a new coefficient α/n_e. The present form of equation (4.13), however, is consistent with the conventional definition of the recombination coefficient.

Although the evaluation of the recombination coefficient α from equation (4.14) and the solution of equations (4.10) for a_k is straightforward, a simpler formulation of three-body recombination would clearly be useful in practical applications. Such a solution has been obtained by means of approximate arguments by Hinnov and Hirschberg (1962). They assume that there exists a certain "critical level," say the qth level, which is such that the probability that an electron which recombines into a higher level will subsequently be de-excited to the ground level is equal to the probability that an electron which recombines into a lower level will subsequently be reionized. That is, since $-\dot{n}_e = \dot{n}_1$ Hinnov and Hirschberg focus attention on whether an electron which recombines into any level will be reionized before it is de-excited to the ground level. The qth level is such that the "loss" to the recombination rate of electrons which recombine into lower levels but are reionized before reaching the ground level is just compensated by electrons which reach the ground level via recombination into higher levels. The net recombination rate is then equal to the rate at which electrons recombine into the qth level and all lower levels, that is

$$n_e^2\alpha = n_e^3 \sum_{1\le k\le q} {}^{\lambda}S_k. \tag{4.15}$$

We note that the same result would follow from our previous analysis [equations (4.14)] if we took

$$a_k = \begin{cases} 0, & 1 \le k \le q \\ \left(\dfrac{n_k}{n_e^2}\right)^*, & k > q \end{cases}$$

and replaced λ' *by* λ. When used in equation (4.12), these values of a_k would then yield

$$\frac{n_k}{n_1} = \left(\frac{n_k}{n_1}\right)^* \quad \text{for} \quad 1 \le k \le q$$

and (4.16)

$$\frac{n_k}{n_e^2} = \left(\frac{n_k}{n_e^2}\right)^* \quad \text{for} \quad k > q.$$

That is, the recombination rate in equation (4.13) would equal that of Hinnov and Hirschberg if it were assumed that the qth and all lower levels are (precisely) in equilibrium with the ground level and that all higher levels are (precisely) in equilibrium with the free electrons (see Exercise 4.3). We note, however, that this is only a sufficient condition for such agreement and that equations (4.16) differ in detail from the physical basis of the Hinnov and Hirschberg argument.

Hinnov and Hirschberg select the critical qth level by the criterion that

$${}^qS_{q-1} = {}_{q-1}S^q \simeq {}_qS^{q+1},$$

where the indicated approximation applies to high-lying levels. By equation (4.7a) or (2.15), the equality ${}^qS_{q-1} = {}_{q-1}S^q$ will be satisfied when

$$\frac{g_q}{g_{q-1}} e^{-\epsilon_{q,q-1}/kT_e} = 1. \tag{4.17}$$

Thus, the qth level is that level where the upward collisional rate $n_e n_q {}_qS^{q+1}$ is approximately equal to the downward rate $n_e n_q {}^qS_{q-1}$. For $k < q$ the downward rate will generally be the greater, whereas for $k > q$ the upward rate will be greater. Assuming a hydrogenic structure for the high-lying levels of the excited atom [see equation (II 4.8)], equation (4.17) becomes

$$\frac{(2N_q)^2}{[2(N_q-1)]^2} \times \exp\left\{-\frac{13.6}{kT_e}\left[\frac{1}{(N_q-1)^2} - \frac{1}{N_q^2}\right]\right\} = 1,$$

where N_q is the principal quantum number of the qth level and kT_e is expressed in electron volts. For $N_q \gg 1$, this yields (see Exercise 4.2)

$$N_q^2 \simeq \frac{13.6}{kT_e}$$

and therefore

$$\epsilon_{q\lambda} = \frac{13.6}{N_q^2} \simeq kT_e. \tag{4.18}$$

With the qth level now specified, Hinnov and Hirschberg evaluate ${}_kS^\lambda$ from equation (2.9) by use of the Thomson ionization cross section (see Chapter V, Sec. 8)

$$Q^{k\to\lambda} = \left(\frac{e^2}{4\pi\epsilon_0}\right)^2 \frac{\pi}{\epsilon}\left(\frac{1}{\epsilon_{k\lambda}} - \frac{1}{\epsilon}\right). \tag{4.19}$$

The rate integrals ${}^\lambda S_k$ which appear in the expression (4.15) for the recombination coefficient α can then be found from the values ${}_kS^\lambda$ by means of the detailed-balancing relation ${}^\lambda S_k/{}_kS^\lambda = (n_k/n_e^2)^*$. With the use of certain numerical approximations consistent with $N_q \gg 1$ and the hydrogenic approximation for the structure for the excited atom, Hinnov and Hirschberg then calculate from equation (4.15) the recombination coefficient α in equation (4.13). For $T_e \lesssim 3000°\text{K}$ they find that α is given approximately by the widely quoted formula

$$\alpha \simeq 5.6 \times 10^{-27} n_e (kT_e)^{-9/2} \text{ cm}^3/\text{sec}, \tag{4.20a}$$

where kT_e is again expressed in electron volts and n_e is in cm^{-3}. If T_e is expressed in °K, equation (4.20a) becomes

$$\alpha \simeq 3.4 \times 10^{-22} n_e \left(\frac{T_e}{1000}\right)^{-9/2} \text{ cm}^3/\text{sec}. \tag{4.20b}$$

When used in the rate equation (4.13) this form for α yields a relatively simple, although approximate, expression for the term $\dot{n}_e$ in the electron continuity equation (4.4).

In the calculations of Hinnov and Hirschberg, the levels in the vicinity of the qth level make the major contribution to the sum (4.15). This suggests that our previous restrictions (3.20) on the relative magnitudes of radiative and collisional processes and, also, considerations of the validity of the hydrogenic approximation need be applied only to the high-lying levels in the vicinity of the qth level. This situation is indicative of the existence of a relatively small group of levels, such that transitions between these levels limit the rate of the recombination process. Calculations

based on a rate-limited model of this kind have been performed, for example, by Byron, et al. (1963, 1962) and by Dugan (1964).

When both radiative and collisional recombination processes are important, the situation is more complicated. To the extent that radiative escape causes departures from the Saha equation in the steady state, when $\dot{n}_e = 0$, it is clear that a macroscopic detailed-balancing relation does not apply to $\dot{n}_e$. Accordingly, the limiting, steady-state form for $\dot{n}_e$ must be modified from that of equation (4.13). For this case of coupled radiative and collisional processes, $\dot{n}_e$ can be evaluated from equations (4.5) and (4.6). Detailed calculations, based on essentially the same formulation, have been performed by Bates, Kingston, and McWhirter (1962). We shall now summarize certain of the results of these calculations.

Bates, Kingston, and McWhirter (1962) have solved numerically the rate equations for $\dot{n}_e$ and $\dot{n}_k$ for a plasma which consists of electrons and hydrogen ions. Although they allow for multiply charged ions, for simplicity, we shall restrict our discussion as before to the case where the ionic charge is unity. They consider two situations, that of optically thin plasmas for which all radiation escapes and β_{kj} and $\beta_{\lambda k} = 1$, and that of plasmas which are optically thick for radiation in the Lyman series, that is, resonance transitions which terminate in the ground level, so that $\beta_{k1} = 0$. In the latter case they consider the following four possibilities for the other radiative processes: $\beta_{kj} = 1$, $\beta_{\lambda k} = 1$; $\beta_{kj} = 1$, $\beta_{\lambda 1} = 0$, $\beta_{\lambda j} = 1$; $\beta_{kj} = 0$, $\beta_{\lambda k} = 1$; and $\beta_{kj} = 0$, $\beta_{\lambda 1} = 0$, $\beta_{\lambda j} = 1$—where here $j > 1$, $k \geq 1$.

As we have already noted, the formulation of Bates, Kingston, and McWhirter is equivalent to our equations (4.5) and (4.6) for $\dot{n}_e$ and $\dot{n}_{k>1} = 0$. They, however, chose to work with the rate equation for $\dot{n}_1$ rather than the rate equation for $\dot{n}_e$. Since $\dot{n}_1 = -\dot{n}_e$ when $\dot{n}_{k>1} = 0$, the equation for $\dot{n}_1$ can be derived from our equations (4.5) and (4.6). Thus we have (see Exercise 4.4)

$$\dot{n}_1 = -\dot{n}_e = -n_1 n_e \left(\sum_{1<l<p} {}_1S^l + {}_1S^{\lambda'} \right) + \sum_{1<l<p} n_l (n_e {}^lS_1 + A_{l1}\beta_{l1}) + n_e^2 (n_e {}^{\lambda'}S_1 + A_{\lambda' 1}\beta_{\lambda' 1}). \tag{4.21}$$

As a result of the fact that equations (4.6) and (4.21) are linear in the ground-level density n_1, the solution to these equations can be written in the form

$$\dot{n}_1 = -\dot{n}_e = n_e^2 \gamma = n_e^2 (\alpha - S n_1/n_e), \tag{4.22}$$

Here γ, α, and S are defined by these relations, which follow the notation of Bates, et al. The quantities α and S depend on n_e and T_e but not n_1. It is important to note that the powers of n_e which appear explicitly in equations (4.22) are purely arbitrary, since γ, α, and S are all functions of

n_e. In particular, the form of equation (4.22) does *not* imply that two-body recombination is dominant.

Bates, Kingston, and McWhirter have solved numerically equations (4.6) and (4.21) to obtain the coefficients α and S in equations (4.22) for the radiation-escape situations indicated previously and for a wide range of n_e and T_e. These calculations are based on the use of the semiclassical cross sections of Gryzinski (1959) (see Chapter II, Sec. 4) to obtain the rate integrals ${}_kS^{\lambda}$ and ${}_kS^{l}$. Although there is some question as to the accuracy of these cross sections, Bates, Kingston, and McWhirter conclude from a comparison with experimental data that the rate integrals calculated therefrom are correct to within a factor of three.

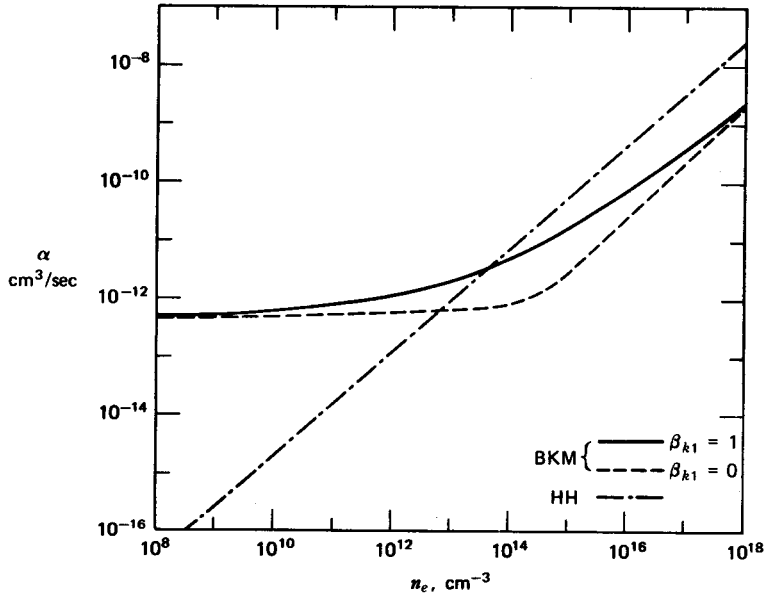


Figure 4. The recombination coefficient α for a hydrogen plasma at $T_e = 8000°$K.

As an example of the results of these calculations, Fig. 4 shows the recombination coefficient α as a function of electron density for a hydrogen plasma at $T_e = 8000°$K. Two cases are given, one for which all radiation escapes, that is all β's are unity, and the other for which the radiation of the resonance Lyman lines is completely absorbed, that is $\beta_{k1} = 0$, but all other radiation escapes so that all other β's are unity. The range of

electron density shown is such that both limits of collision-dominated recombination and radiation-dominated recombination are approached for high and low n_e, respectively. In the former case α is proportional to n_e, whereas in the latter case α is independent of n_e. Also shown in Fig. 4 is the Hinnov and Hirschberg value of the recombination coefficient from equations (4.22). (Note, however, that in Fig. 4 the Hinnov and Hirschberg formula is applied above the intended limit of 3000°K.) In Fig. 5 are

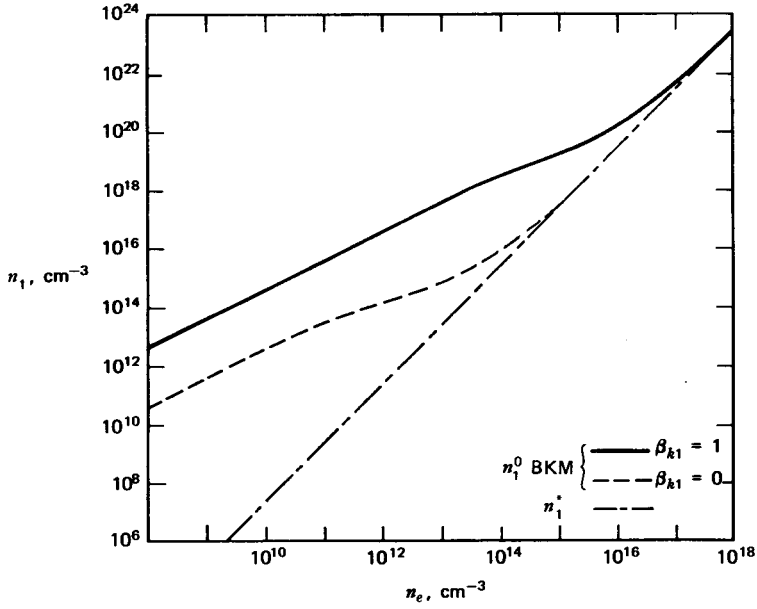


Figure 5. The steady-state (n_1^0) and equilibrium (n_1^*) ground-level densities for a hydrogen plasma at $T_e = 8000$°K.

given the values n_1^0 of the ground-level density in steady state for the two cases of Fig. 4 and the equilibrium value $n_1^* = (n_1/n_e^2)^* n_e^2$ of the ground-level density. The ionization coefficient S in equation (4.22) can be readily found in terms of α and n_1^0 from the relation

$$S = \frac{n_e}{n_1^0}\alpha. \tag{4.23}$$

Figs. 4 and 5 show not only the effect of different degrees of radiation escape on the recombination coefficient and the accuracy of the Hinnov and Hirschberg formula in comparison with the more exact calculations of Bates, et al., but also the effect of radiation escape on the steady-state densities. In this context we note that a larger value of n_1^0 for a given n_e

is equivalent to a smaller value of n_e for a given n_1 at steady state. Thus, since $n_1^0 > n_1^*$, we have at steady state $n_e^2/n_1 < (n_e^2/n_1)^*$, as discussed in the previous section.

It is also of interest to examine the populations n_k of the various levels, which are calculated from equations (4.6) and which through equations (4.21) and (4.22) are required to obtain results such as those of Figs. 4 and 5. Values of n_k at $T_e = 8000°K$ for several electron densities as given by Bates and Kingston (1963) are shown in Fig. 6, for the case of complete

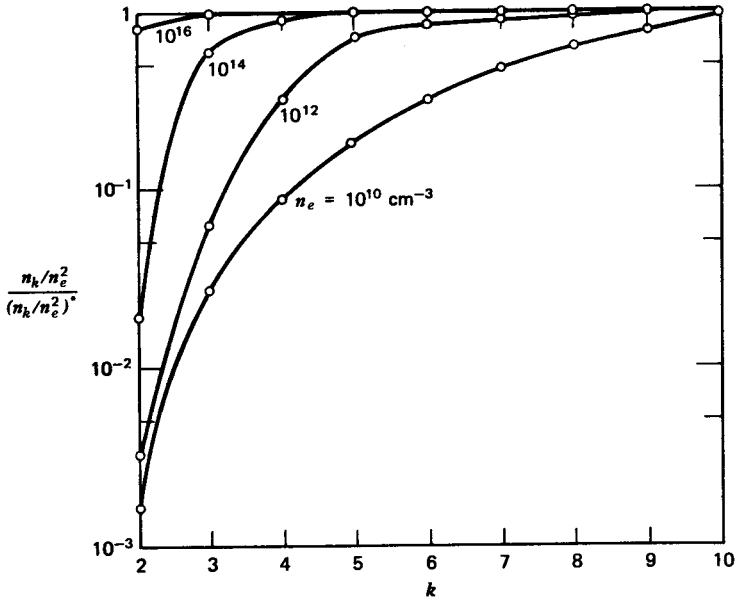


Figure 6. Populations n_k of the various excited levels in comparison with their values n_k^* in equilibrium with the free electrons for an optically thin hydrogen plasma at $T_e = 8000°K$ under conditions such that $n_e^2/n_1 \geq (n_e^2/n_1)^*$.

radiation escape. To simplify the presentation the values shown apply to the situation where $n_e^2/n_1 \geq (n_e^2/n_1)^*$ as would be the case for a recombining plasma. It can be seen from this figure that the assumption that the high-lying levels are in equilibrium with the free electrons is verified.

For the case of complete radiation escape the formulation of Bates, et al., which yields the results of Figs. 4, 5, and 6, is completely in accord with the preceding discussion of this section. When the radiation of the Lyman lines is absorbed, on the other hand, they allow in their calculations for the possibility that $\dot{n}_2 \neq 0$. The results of these calculations show, however, that for most cases $\dot{n}_2 \to 0$ in a time very much less than that required for

$\dot{n}_e \to 0$. Under these conditions their results, in particular those shown in Fig. 4, reduce to those that would have been obtained if $\dot{n}_2$ had been set equal to zero initially.

We conclude this section with a comparison, shown in Fig. 7, of the recombination coefficients calculated by Bates, et al. and by Hinnov and

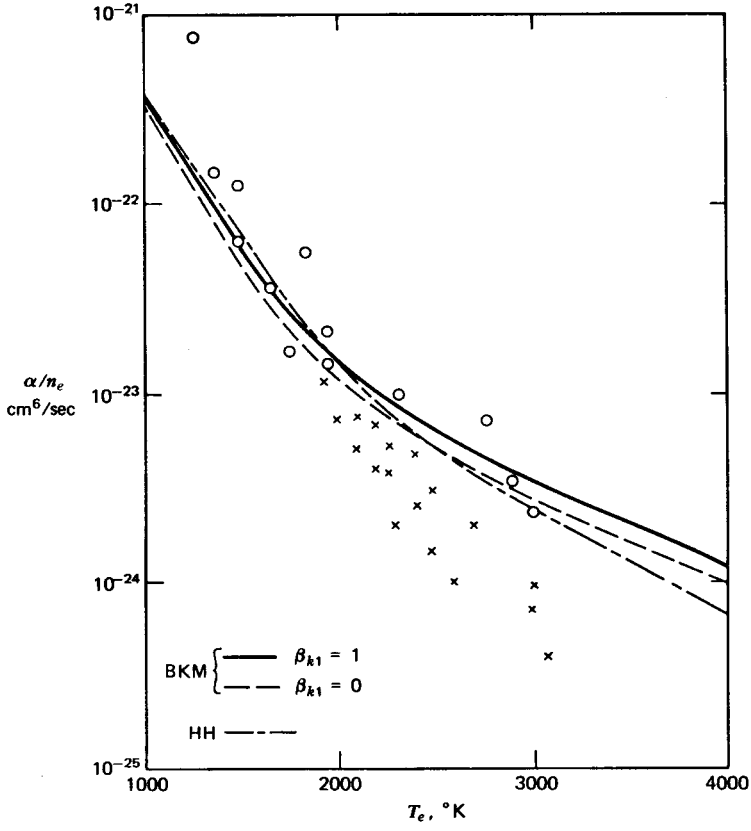


Figure 7. Comparison of the recombination coefficient α/n_e calculated by Bates, Kingston, and McWhirter (1962) and by Hinnov and Hirschberg (1962) with the experimental data of Aleskovskii (1963), O, and Cool and Zukoski (1966), x.

Hirschberg with experimental data for cesium and potassium plasmas measured by Aleskovskii (1963) and by Cool and Zukoski (1966). (The values shown here for Bates, et al. are based on $n_e = 10^{13}$ cm^{-3}, which corresponds approximately to n_e in the experiments. The calculated values of α/n_e do not vary greatly with n_e, particularly for $T_e \leq 2000$°K, nor do

they differ significantly from the values for a "pseudoalkali" plasma given in Table 4A of Part I of their paper.) It must be recognized that although the theoretical calculations are based on hydrogenic atoms, the experimental data are for alkali-metal plasmas. In view of the differences in structure between the alkali-metal atoms and hydrogen and the differences between the two sets of data, the comparison can be regarded as reasonable.

The discussion and results of this section have been based on the assumption that the free-electron distribution function can be taken as Maxwellian. In the following section we consider the more general case in which the Boltzmann equation and the rate equations are investigated simultaneously.

Exercise 4.1. Find the expression for $\dot{n}_e$ for use in the electron continuity equation (4.4) that applies to the two-level model of the previous section. Express your result solely in terms of n_e, n_1, the rate integrals, such as ${}_1S^{\lambda}$, which depend on T_e, and the radiation-escape parameters. Show that when radiative processes can be neglected, your result can be written in the form of equation (4.13), so that $n_e^2/n_1 = (n_e^2/n_1)^*$ when $\dot{n}_e = 0$.

Exercise 4.2. With the use of numerical approximations consistent with the condition $N_q \gg 1$, derive equation (4.18) of the Hinnov and Hirschberg analysis. Show that these approximations are justified for $T_e \lesssim 3000°\text{K}$.

Exercise 4.3. Show that regardless of equations (4.16), equation (4.15) is *not* obtained by setting $p = q + 1$ in equation (4.9), unless λ' is replaced by λ. Explain this apparent contradiction with the use of Exercise 3.5 and the two-level model.

Exercise 4.4. Derive equation (4.21) for $\dot{n}_1$, (a) directly from physical considerations, and (b) from equations (4.5) and (4.6) for $\dot{n}_e$ and $\dot{n}_{k>1}$.

5. THE EFFECTS OF A NON-MAXWELLIAN DISTRIBUTION FUNCTION

In our discussion of ionizational nonequilibrium, we have assumed up to this point that as a result of electron-electron collisions the free electron distribution function differs only slightly from a local, isotropic Maxwellian function. Accordingly, our formulation of the rate equations for $\dot{n}_e$ and $\dot{n}_k$ has been uncoupled from the Boltzmann equation for f^0. The electron distribution function in a collision-dominated plasma will be Maxwellian if the conditions of Chapter VIII, Sec. 3, hold for the electrons that participate in exciting and ionizing collisions as well as for thermal electrons. Here we

shall continue to assume that the conditions on the spatial gradients, time variations, and electric fields, and on the relative magnitudes of the electron-electron and electron/heavy-particle elastic collisions frequencies [i.e., $\bar{\nu}_{ee} \gg \sum_h (m_e/m_h)\bar{\nu}_{eh}$] are satisfied as before. We shall, however, relax our previous restriction, $\bar{\nu}_{ee} \gg \bar{\nu}_{NE}$, on the nonelastic collision frequency. Under these circumstances the departure of f_e from an isotropic function f^0 will still be small and f^0 will be determined by the interplay between electron-electron elastic collisions and electron/heavy-particle nonelastic collisions. Consequently, we must now consider the coupled problem in which the appropriate form of the Boltzmann equation for f^0 and the rate equations for $\dot{n}_e$ and $\dot{n}_k$ are solved simultaneously. In this section, then, we shall discuss the effects of the non-Maxwellian distribution function that results from the solution of these coupled equations on departures from the Saha equation and on the populations of the excited levels at steady state and on ionization and recombination rates. Since these effects occur at relatively low electron densities, it is of particular interest to find that the electron density above which f^0 remains Maxwellian and our results of Secs. 3 and 4 continue to apply. Our discussion of these matters follows closely the work of Shaw, Mitchner, and Kruger (1970).

Under the conditions specified in the preceding paragraph, the distribution function f^0 is determined by the first Cartesian tensor equation, equation (VII 6.40), with all terms except $\mathbb{C}^0_{ee}$ omitted as relatively small in comparison with $\mathbb{C}^0_{ee}$ [as in equation (VIII 3.2)] and with the addition of the appropriate nonelastic collision term. If we denote the nonelastic collision term by $\mathbb{C}^0_{NE}$, this equation for f^0 can be written symbolically as

$$\mathbb{C}^0_{ee} + \mathbb{C}^0_{NE} = 0, \tag{5.1}$$

where $\mathbb{C}^0_{ee}$ is given by equation (VII 6.32). The general expression for $\mathbb{C}^0_{NE}$ involves a complicated summation over the various non-elastic collisional processes discussed in Sec. 2. When the mean electron thermal energy $(3/2)kT_e$ is much less than the excitation energy ϵ_{12} of the first excited level, however, Shaw, Mitchner, and Kruger find that for many plasmas the dominant contribution to $\mathbb{C}^0_{NE}$ is associated with collisional excitation and de-excitation of the first excited level from and to the ground level. For simplicity we shall take this to be the case, since the effects in question are adequately illustrated under these circumstances.

The term $\mathbb{C}^0_{NE}$ in equation (5.1) can be formulated most directly as in Chapter VIII, Sec. 8, on the basis that equations (VII 6.39) and (5.1) are speed-dependent energy equations for the free electrons. That is, $\mathbb{C}^0_{NE}(C)$ when multiplied by the factor $\frac{1}{2}m_e C^2 4\pi C^2\, dC = \epsilon(4\pi/m_e)\sqrt{2\epsilon/m_e}\, d\epsilon$ represents the net rate of energy gain per unit volume as a result of nonelastic collisions for electrons in the speed class dC. Therefore, for electrons of

energy $\epsilon = \frac{1}{2}m_e C^2$ greater than ϵ_{12} the contribution of nonelastic collisions of the type $e + A_1 \rightleftharpoons e + A_2$ to $\mathbb{C}^0_{NE}$ is given by

$$\mathbb{C}^0_{NE}\epsilon\frac{4\pi}{m_e}\sqrt{\frac{2\epsilon}{m_e}}\,d\epsilon = -n_e n_1 f^0(\epsilon)\sqrt{\frac{2\epsilon}{m_e}}\,Q^{1\to 2}(\epsilon)\times\epsilon\frac{4\pi}{m_e}\sqrt{\frac{2\epsilon}{m_e}}\,d\epsilon$$

$$+ n_e n_2 f^0(\epsilon-\epsilon_{12})\sqrt{\frac{2}{m_e}(\epsilon-\epsilon_{12})}\,Q^{2\to 1}(\epsilon-\epsilon_{12})$$

$$\times\epsilon\frac{4\pi}{m_e}\sqrt{\frac{2}{m_e}(\epsilon-\epsilon_{12})}\,d(\epsilon-\epsilon_{12}). \tag{5.2}$$

Here, as in equations (2.3) and (2.4), the relative speed in the nonelastic collision frequency has been written as $\sqrt{2\epsilon/m}$ and the differential volume element $4\pi C^2\,dC$ in velocity space as $(4\pi/m)\sqrt{2\epsilon/m_e}\,d\epsilon$. In equation (5.2) the first term on the right-hand side represents the loss of energy in the amount ϵ for electrons of the class $d\epsilon$ as a result of inelastic exciting collisions of the type $e + A_1 \to e + A_2$. The second term represents the gain of energy of the amount ϵ for electrons of the same class as a result of inverse, superelastic collisions of the type $e + A_2 \to e + A_1$. In the inverse collisions the energy of the electron is $\epsilon - \epsilon_{12}$ before the collision and ϵ after the collision. The nonelastic cross sections $Q^{1\to 2}(\epsilon)$ and $Q^{2\to 1}(\epsilon-\epsilon_{12})$ are related by detailed balancing as follows:

$$f_M(\epsilon)\sqrt{\frac{2\epsilon}{m_e}}\,Q^{1\to 2}(\epsilon)\frac{4\pi}{m_e}\sqrt{\frac{2\epsilon}{m_e}} = \left(\frac{n_2}{n_1}\right)^* f_M(\epsilon-\epsilon_{12})\sqrt{\frac{2}{m_e}(\epsilon-\epsilon_{12})}$$

$$\times\,Q^{2\to 1}(\epsilon-\epsilon_{12})\frac{4\pi}{m_e}\sqrt{\frac{2}{m_e}(\epsilon-\epsilon_{12})}. \tag{5.3}$$

With $(n_2/n_1)^*$ given by equation (2.6) and the Maxwellian distribution $f_M(\epsilon)$ given by

$$f_M(\epsilon) = \left(\frac{m_e}{2\pi kT_e}\right)^{3/2} e^{-\epsilon/kT_e} \tag{5.4}$$

equations (5.2) and (5.3) yield for the nonelastic-collision term in question

$$\mathbb{C}^0_{NE} = n_e\left[-n_1 f^0(\epsilon) + \frac{g_1}{g_2}n_2 f^0(\epsilon-\epsilon_{12})\right]\sqrt{\frac{2\epsilon}{m_e}}\,Q^{1\to 2}(\epsilon), \tag{5.5}$$

where g_1 and g_2 are the degeneracies of the two levels.

We note that equation (5.5) accounts only for the effect of nonelastic $e + A_1 \rightleftharpoons e + A_2$ collisions on electrons of energy greater than ϵ_{12}. More precisely, we have considered only the electron *before* an inelastic $e + A_1 \rightarrow e + A_2$ collision and the electron *after* a superelastic $e + A_2 \rightarrow e + A_1$ collision, both of which must have energies greater than ϵ_{12}. In general there are also nonelastic effects associated with the electrons after an inelastic collision and before a superelastic collision, which can have any energy. These latter electrons, however, will be primarily of thermal energies. For such thermal electrons under conditions of interest here, with $kT_e \ll \epsilon_{12}$, nonelastic collisions can usually be neglected in comparison with elastic electron-electron collisions. Therefore we shall not include these effects in our model, except as noted following equations (5.13).

We can now write out in full the Boltzmann equation (5.1) for f^0 with equation (VII 6.32) for $\mathbb{C}_{ee}^0$ and equation (5.5) for $\mathbb{C}_{NE}^0$. We have, after expressing $\mathbb{C}_{ee}^0$ in terms of ϵ and rearranging,

$$\epsilon \frac{d}{d\epsilon}\left[\epsilon a(\epsilon)\frac{df^0}{d\epsilon}(\epsilon) + b(\epsilon)f^0(\epsilon)\right] = c(\epsilon)\left[f^0(\epsilon) - \frac{g_1 n_2}{g_2 n_1} f^0(\epsilon - \epsilon_{12})\right], \quad (5.6)$$

where the dimensionless quantities $a(\epsilon)$, $b(\epsilon)$, and $c(\epsilon)$ are defined by

$$a(\epsilon) \equiv \frac{2}{3}\left[\frac{1}{\epsilon}\int_0^{\epsilon} \epsilon f^0 \frac{4\pi}{m_e}\sqrt{\frac{2\epsilon}{m_e}}\, d\epsilon + \sqrt{\epsilon}\int_{\epsilon}^{\infty} \frac{1}{\sqrt{\epsilon}} f^0 \frac{4\pi}{m_e}\sqrt{\frac{2\epsilon}{m_e}}\, d\epsilon\right], \quad (5.7a)$$

$$b(\epsilon) \equiv \int_0^{\epsilon} f^0 \frac{4\pi}{m_e}\sqrt{\frac{2\epsilon}{m_e}}\, d\epsilon, \quad (5.7b)$$

$$c(\epsilon) \equiv \epsilon^2 \frac{n_1 Q^{1\rightarrow 2}(\epsilon)}{n_e m_e^2 \Gamma_{ee}}. \quad (5.7c)$$

Since Γ_{ee} is given by equation (VII 4.14) as

$$\Gamma_{ee} = \frac{g^4}{4} Q_{ee}^{(1)}, \quad (5.8)$$

it can be seen that the quantity $c(\epsilon)$ expresses the ratio of the electron nonelastic excitation collision frequency to the electron-electron elastic collision frequency. Accordingly, this quantity is a measure of the relative importance of nonelastic collisions.

Equation (5.6) for f^0 has been solved simultaneously with the rate equations for n_2 and n_e numerically by Shaw, Mitchner, and Kruger. Before discussing the results of these calculations, we shall consider an approximate solution to equation (5.6) which will serve to establish the qualitative effects of nonelastic collisions on the distribution function. We

first note that since the nonelastic cross section $Q^{1\to 2}(\epsilon)$ is zero for $\epsilon < \epsilon_{12}$, the quantity $c(\epsilon)$ vanishes for energies below the threshold energy ϵ_{12}. Therefore, for $\epsilon < \epsilon_{12}$ the right-hand side of equation (5.6) is zero and the equation reduces to

$$\mathbb{C}_{ee}^{0} = 0. \qquad (\epsilon \leq \epsilon_{12}) \tag{5.9}$$

Accordingly, as in Chapter VIII, Sec. 3, the distribution function f^0 will be nearly Maxwellian for electrons below threshold energy. We now examine the expressions (5.7a) and (5.7b) for $a(\epsilon)$ and $b(\epsilon)$. Since we have assumed that $kT_e \ll \epsilon_{12}$, for energies greater than ϵ_{12} the second integral in equation (5.7a) will make only a small contribution to $a(\epsilon)$ and, in the remaining integral and in the expression for $b(\epsilon)$, the primary contribution will come from those energies for which $f^0 \simeq f_M$, at least as long as f^0 $(\epsilon \geq \epsilon_{12})$ is not much larger than f_M $(\epsilon \geq \epsilon_{12})$. It follows that the upper limit ϵ in the remaining two integrals in equations (5.7a) and (5.7b) can be replaced, approximately, by ∞, so that we have

$$a(\epsilon) \simeq \frac{2/3}{\epsilon}\int_0^\infty \epsilon f^0 \frac{4\pi}{m_e}\sqrt{\frac{2\epsilon}{m_e}}\,d\epsilon = \frac{2/3}{\epsilon}\bar{\epsilon}$$

$$= \frac{1}{\epsilon}kT_e \tag{5.10}$$

and

$$b(\epsilon) \simeq \int_0^\infty f^0 \frac{4\pi}{m_e}\sqrt{\frac{2\epsilon}{m_e}}\,d\epsilon = 1,$$

for $\epsilon \geq \epsilon_{12}$. With the use of these values, equation (5.6) reduces to

$$\epsilon\frac{d}{d\epsilon}\left[kT_e\frac{df^0(\epsilon)}{d\epsilon} + f^0(\epsilon)\right] = c(\epsilon)\left[f(\epsilon) - \frac{g_1 n_2}{g_2 n_1}f^0(\epsilon - \epsilon_{12})\right] \qquad (\epsilon \geq \epsilon_{12}) \tag{5.11}$$

Although our arguments which led to the approximations (5.10) for $a(\epsilon)$ and $b(\epsilon)$ apply only to high-energy $(\epsilon \geq \epsilon_{12})$ electrons, it can be seen that equation (5.11) is satisfied by a Maxwellian distribution also at thermal energies for which $\epsilon < \epsilon_{12}$ and $c(\epsilon) = 0$.

Equations (5.9) and (5.11) serve as a good approximation to the Boltzmann equation (5.6) for f^0. Analytic solutions to these equations have been obtained by Shaw, Mitchner, and Kruger for several approximate representations of the quantity $c(\epsilon)$. The solution of equation (5.11) was obtained by means of LaPlace-transform methods; here care must be exercised with

respect to boundary conditions and exponentially growing solutions. The simplest such solution is for the case where $c(\epsilon)/\epsilon$ is taken to be constant, say c_0/kT_e, for $\epsilon \geq \epsilon_{12}$, that is,

$$\frac{c(\epsilon)}{\epsilon} = \begin{cases} 0 & \text{for} \quad \epsilon < \epsilon_{12} \\ \dfrac{c_0}{kT_e} & \text{for} \quad \epsilon \geq \epsilon_{12} \end{cases}. \tag{5.12}$$

By comparison with equation (5.7c) it can be seen that this corresponds to an excitation cross section $Q^{1\to 2}(\epsilon)$ which decreases as $1/\epsilon$ from its value at threshold. Although this is not a particularly good representation of the behavior of actual cross sections, the results are in qualitative agreement with the accurate numerical solutions and, in fact, the value of c_0 can be adjusted to give satisfactory quantitative values as well.

With equation (5.12) for $c(\epsilon)$, the solution to equations (5.9) and (5.11) for f^0 is found to be

$$f^0(\epsilon) = \left(\frac{m_e}{2\pi kT_e}\right)^{3/2} [e^{-\epsilon/kT_e} - (1 - P_2)(1 - \alpha_1)e^{-\epsilon_{12}/kT_e}], \quad \text{for} \quad \epsilon \leq \epsilon_{12}, \tag{5.13a}$$

and

$$f^0(\epsilon) = \left(\frac{m_e}{2\pi kT_e}\right)^{3/2} e^{-\epsilon_{12}/kT_e} \Bigg\{ P_2\, e^{-(\epsilon-\epsilon_{12})/kT_e} + \alpha_1(1 - P_2)e^{-(\epsilon-\epsilon_{12})/\alpha_1 kT_e}$$

$$+ P_2(1 - P_2)e^{-\epsilon_{12}/kT_e} \left[(\alpha_1 - 1) + \left(\frac{\alpha_1^2}{\alpha_1 + \alpha_2} - P_2(1 - \alpha_1) \right) \right.$$

$$\left. \times\, e^{(\epsilon - 2\epsilon_{12})/\alpha_2 kT_e} \right] \Bigg\} \quad \text{for} \quad \epsilon_{12} \leq \epsilon \leq 2\epsilon_{12}. \tag{5.13b}$$

Here the parameters P_2, α_1, and α_2 are defined by

$$P_2 \equiv \frac{n_2/n_1}{(n_2/n_1)^*},$$

$$\frac{1}{\alpha_1} \equiv \sqrt{c_0 + \frac{1}{4}} + \frac{1}{2}, \tag{5.13c}$$

and

$$\frac{1}{\alpha_2} \equiv \sqrt{c_0 + \frac{1}{4}} - \frac{1}{2}.$$

An expression similar to equation (5.13b), but more complicated in form, is obtained for the energy range $2\epsilon_{12} \leq \epsilon \leq 3\epsilon_{12}$, and so on for even higher energies.

As we had indicated previously, the solution (5.13a) for $\epsilon \leq \epsilon_{12}$ is very nearly a Maxwellian distribution. The second term in the brackets in equation (5.13a) is a constant which is usually set equal to zero because of the condition that $f^0(\epsilon) \to 0$ as $\epsilon \to \infty$. This term reflects the coupling of the two solutions at $\epsilon = \epsilon_{12}$ and is necessary in order that continuity of f^0 and $df^0/d\epsilon$ is maintained at that point. This solution (5.13a) of equation (5.9) is based on arguments similar to those that led to equation (5.11). In addition, there is a possible singularity of the solution as $\epsilon \to 0$. This point is discussed in relation to the effect of nonelastic collisions on low-energy thermal electrons by Shaw, Mitchner and Kruger (1970), Part I.

The use in the solution (5.13) of the normalization factor $(m_e/2\pi kT_e)^{3/2}$ and the factor $1/kT_e$ in the exponentials is consistent with our assumption that $kT_e \ll \epsilon_{12}$. When this holds, the dominant contribution to the integrals

$$\int_{-\infty}^{\infty} f^0 \, d^3C = 1$$

and

$$\tfrac{3}{2}kT_e \equiv \tfrac{1}{2}m_e \overline{C^2} = \int_{-\infty}^{\infty} \tfrac{1}{2}m_e C^2 f^0 \, d^3C$$

comes from the first term in brackets in the part (5.13a) of the solution.

The approximate solution (5.13) for f^0 is shown in Fig. 8 in terms of the two parameters P_2 and c_0, for $kT_e = 0.1\epsilon_{12}$. When $n_2/n_1 = (n_2/n_1)^*$ we have $P = 1$ and the distribution function is Maxwellian regardless of the value of c_0. For $n_2/n_1 < (n_2/n_1)^*$ and hence $P_2 < 1$, the high energy "tail" of the distribution is depressed below its Maxwellian value. This depression increases with increasing c_0 and therefore, since $c_0 \propto n_e^{-1}$, becomes more important as the electron density decreases. Physically, the depression of the tail of f^0 when $P_2 < 1$ results from the fact that since $n_2/n_1 < (n_2/n_1)^*$ the loss of high energy electrons in inelastic collisions of the type $e + A_1 \to e + A_2$ is no longer balanced by the gain of such electrons in superelastic collisions of the type $e + A_2 \to e + A_1$.

The consequences of the solution (5.13) shown in Fig. 8 can be found only when this solution is used in conjunction with an appropriately modified form of the rate equations of Sec. 2. To illustrate these calculations in a simple way, we shall assume for the moment that conditions are such that we can apply the approximate form (3.4) of the rate equation for $\dot{n}_2$. That is, we assume that the population n_2 of the first excited level is determined by a balance between the rates of collisional excitation from the ground

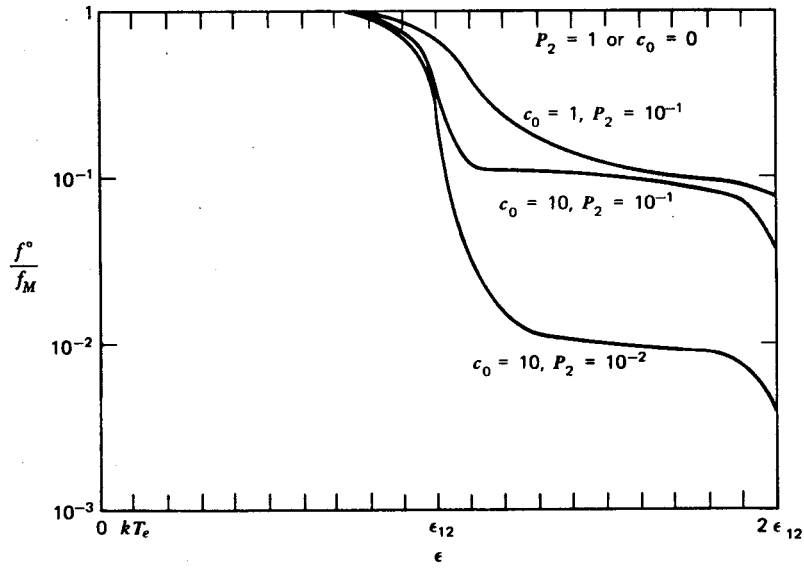


Figure 8. The approximate solution (5.13) for the electron distribution function for the case $kT_e = 0.1\ \epsilon_{12}$.

level and collisional and radiative de-excitation to the ground level. The solution to the corresponding rate equation (3.4) can be written in the form

$$\frac{n_2}{n_1} = \frac{{}_1S^2/{}^2S_1}{1 + R_2}, \tag{5.14}$$

where as before the dimensionless parameter

$$R_2 \equiv \frac{A_{21}\beta_{21}}{n_e\, {}^2S_1} \tag{5.15}$$

is a measure of the importance of radiation escape. In Sec. 3, where we assumed a Maxwellian distribution, we were able to replace ${}_1S^2/{}^2S_1$ by $(n_2/n_1)^*$ with the use of the detailed balancing relation (2.6). Now, however, with the possibility of a non-Maxwellian distribution we can apply detailed balancing only as in equation (5.3) to the relation between the cross sections $Q^{1\to 2}$ and $Q^{2\to 1}$ and not to the relation between the rate integrals ${}_1S^2$ and ${}^2S_1$. At the same time, we note that the distribution (5.13) is nearly Maxwellian for the low-energy electrons that participate in de-exciting collisions of the type $e + A_2 \to e + A_1$; therefore we can regard ${}^2S_1$ as unaffected by a non-Maxwellian distribution. If we then denote by $({}_1S^2)_M$

and $({}^2S_1)_M$ the forms (2.3) and (2.4) of the rate integrals that are obtained with $f^0 = f_M$, we have

$$ {}^2S_1 \simeq ({}^2S_1)_M $$

and

$$ \frac{({}_1S^2)_M}{{}^2S_1} \simeq \frac{({}_1S^2)_M}{({}^2S_1)_M} = \left(\frac{n_2}{n_1}\right)^*. $$

Equation (5.14) for n_2/n_1 can then be written as

$$ \frac{n_2/n_1}{(n_2/n_1)^*} = \frac{{}_1S^2/({}_1S^2)_M}{1 + R_2}. \tag{5.16} $$

The ratio ${}_1S^2/({}_1S^2)_M$ in equation (5.16) can be evaluated by the use in equation (2.3) of the non-Maxwellian distribution (5.13b) in place of f_M. In the present case, where the quantity $c(\epsilon)/\epsilon$ is taken to be a constant, c_0/kT_e, so that $Q^{1\to 2} \propto 1/\epsilon$ for $\epsilon \geq \epsilon_{12}$ [see equations (5.7c) and (5.12)], this integration yields

$$ \frac{{}_1S^2}{({}_1S^2)_M} = P_2 + (1 - P_2)\alpha_1^2 \frac{\left(\frac{\epsilon_{12}}{kT_e} + \alpha_1\right)}{\left(\frac{\epsilon_{12}}{kT_e} + 1\right)} + \text{terms of the order } e^{-\epsilon_{12}/kT_e}, \tag{5.17} $$

where, as before, $P_2 = (n_2/n_1)/(n_2/n_1)^*$ and $\alpha_1^{-1} = (c_0 + 1/4)^{1/2} + 1/2$.

Substitution of equation (5.17) in equation (5.16) gives a single equation for $(n_2/n_1)/(n_2/n_1)^*$, the solution of which is given in Fig. 9 for $kT_e = 0.1\epsilon_{12}$. This solution shows the influence of radiation escape, expressed in terms of the parameter R_2 of equation (5.15), on the population of the first excited level. The importance of nonelastic collisions and hence a non-Maxwellian distribution depends on the parameter c_0, which is a measure of the relative magnitude of the nonelastic excitation collision frequency in comparison with the electron-electron elastic collision frequency. For $c_0 = 0$ the effects of nonelastic collisions are absent and these results reduce to those of equation (3.5) and, for example, Fig. 1, Sec. 3. To relate the results of Fig. 9 to actual plasmas, approximate values of $n_e kT_e c_0/n_1$ are given in Table 2 for conditions that correspond to those of Table 1 of Sec. 3. Here the quantity c_0, which is based on an approximate model of the cross section $Q^{1\to 2}$, has been evaluated from the actual values of $Q^{1\to 2}$ for the various species by setting $Q^{1\to 2}(\epsilon)$ for the model equal to the actual value of $Q^{1\to 2}(\epsilon)$ at $\epsilon = \epsilon_{12} + kT_e$.

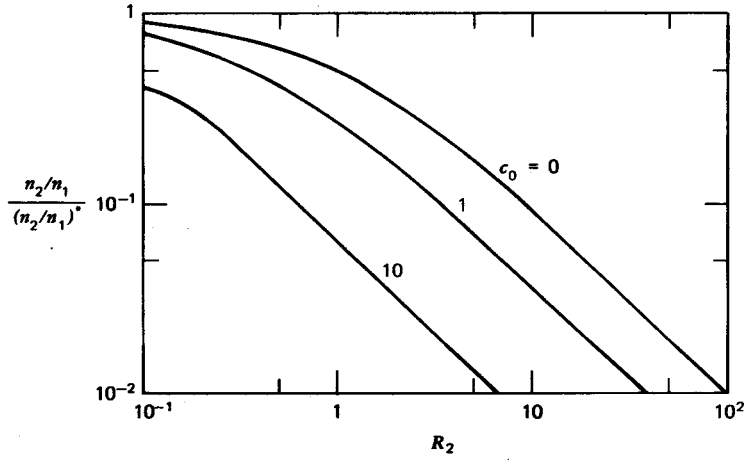


Figure 9. The effect of a non-Maxwellian distribution function on the population of the first excited level for the case $kT_e = 0.1\ \epsilon_{12}$, based on the two-level model and the approximate solution (5.13) for f^0. The parameters R_2 and c_0 are measures of the degree of radiation escape and the relative importance of nonelastic collisions, respectively.

Solution of the approximate form (5.11) of the Boltzmann equation shows qualitatively the effects on the distribution function that result from nonelastic collisions. Furthermore, with a judicious choice of an analytic model for the nonelastic-collision parameter $c(\epsilon)$ it is possible to obtain reasonable quantitative results based on this equation. This can be established from the comparison of such results with the solution of the more exact form (5.6) of the Boltzmann equation. Solutions of the latter equation have been obtained numerically by Shaw, Mitchner, and Kruger for hydrogen and alkali-metal plasmas.

It is of particular interest to examine the effects of a non-Maxwellian distribution function on the results of Bates, Kingston, and McWhirter (1962), which were discussed in the previous section. Shaw, Mitchner, and Kruger have solved the Boltzmann equation (5.6) simultaneously with the rate equations (4.21) and (4.6) for $\dot{n}_e$ and $\dot{n}_k$ using for hydrogen plasmas

Table 2 Approximate Values of the Nonelastic-Collision Parameter c_0 Defined by Equations (5.7c) and (5.12)

Species	Cs	K	Na	A	H
T_e (°K)	2500	2500	2500	10,000	10,000
$n_e kT_e c_0/n_1$	3.92×10^{-3}	1.32×10^{-3}	4.40×10^{-4}	4.17×10^{-5}	2.24×10^{-4}

the same cross sections employed by BKM. Thus by setting f^0 equal to a Maxwellian distribution, and by considering from 16 to 21 levels of the hydrogen atom, they are able to reproduce the results of BKM to within a few percent. On the other hand, when f^0 is found numerically from the solution of the Boltzmann equation, substantial departures from the previous results are observed at low electron densities. In these calculations, the coefficients $a(\epsilon)$ and $b(\epsilon)$ in equation (5.6) have been modified from the forms given by equations (5.7a) and (5.7b) to include in this formulation the effects of electron-atom elastic collisions (see Exercise 5.1). Fig. 10 shows the effects of a non-Maxwellian distribution on the steady-state ($\dot{n}_e = 0$) values of the ratio $(n_e^2/n_1)/(n_e^2/n_1)^*$ as a function of electron density n_e, for

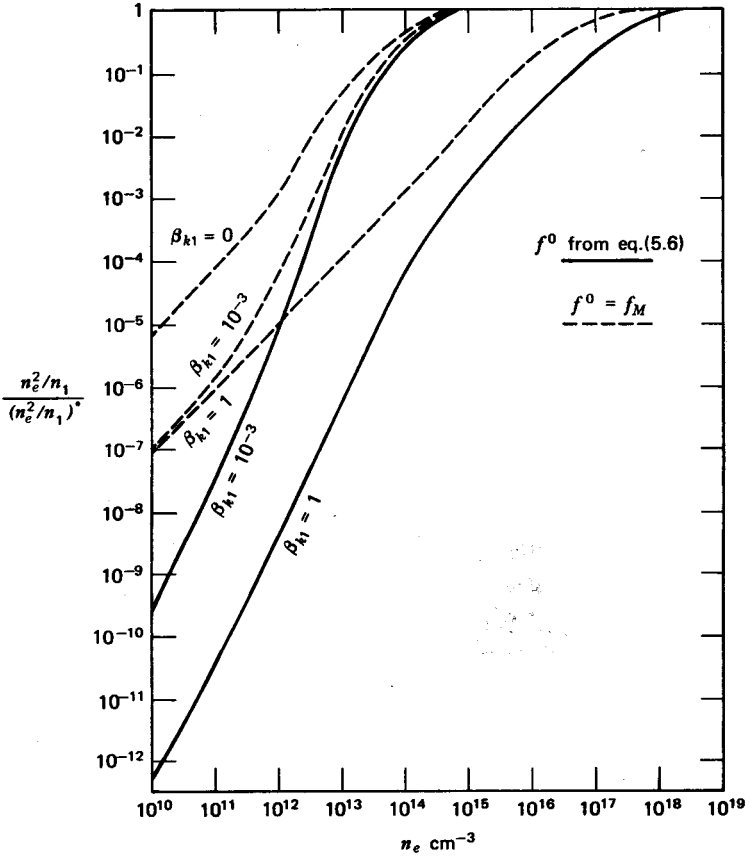


Figure 10. The effect of a non-Maxwellian distribution function on departures from the Saha equation at steady state for hydrogen at 8000°K, as calculated by Shaw, Mitchner, and Kruger (1970).

hydrogen at $T_e = 8000°K$. In these calculations all nonresonance radiation is assumed to escape from the plasma ($\beta_{kj\neq 1} = \beta_{\lambda k} = 1$), and the resonance-radiation escape parameter β_{k1} is taken as a parameter. At the lower values of n_1 and hence of n_e the departures from the Saha equation $[n_e^2/n_1 = (n_e^2/n_1)^*]$ found by Shaw, et al. are seen to be much greater than those predicted by the use of a Maxwellian distribution. Because of this, at the lower electron densities the effects of atom-atom excitation, which were not included in the calculations, may become important.

In addition to the effects on the steady-state values of n_e^2/n_1, a non-Maxwellian distribution also modifies the values of $\dot{n}_e$ in a nonsteady plasma. Bates, Kingston, and McWhirter express $\dot{n}_e$ in the form

$$\dot{n}_e = n_e^2(-\alpha + Sn_1/n_e), \tag{5.18}$$

where α and S are functions of n_e and T_e but not of n_1. It is found by Shaw, et al. that the recombination coefficient α, which depends only on the low (thermal) energy portion of f^0, is essentially unchanged when $f^0 \neq f_M$, since f^0 is in any case nearly Maxwellian for $\epsilon < \epsilon_{12}$. In fact, for hydrogen plasmas the first-level excitation energy $\epsilon_{12} = 10.2$ eV is such a large fraction of the ionization potential $\epsilon_{1\lambda} = 13.6$ eV that the only rate integrals that are affected by $f^0 \neq f_M$ are those for upward transitions from the ground level. The ionization coefficient S is affected, however, by the distribution function. In Fig. 11 the ratio S/S_M of the ionization coefficient with $f^0 \neq f_M$ to that with $f^0 = f_M$ is shown as a function of n_e/n_1 for several values of n_e. Here for the cases of complete escape of resonance radiation ($\beta_{k1} = 1$), the values of S_M are those calculated by BKM with $f^0 = f_M$. Since results for the case of partial escape are not given by BKM, the values of S_M for $\beta_{k1} = 10^{-3}$ are those obtained by Shaw, et al. when f^0 is set equal to f_M. The curves for $\beta_{k1} = 10^{-3}$ in Fig. 10 were obtained similarly. When the resonance radiation is completely trapped $\beta_{k1} = 0$, it is found that the results of Shaw, et al. do not differ significantly from those of BKM for which $f^0 = f_M$, so that in this case $S/S_M \simeq 1$. In the calculations with $f^0 = f_M$, S is independent of n_1. With $f^0 \neq f_M$, however, S is a function of n_1 because of the role of the nonelastic-collision parameter $c(\epsilon)$ in the solution for f^0 [see equation (5.7c)]. In the regions of net ionization for relatively small n_e/n_1, the effect of the non-Maxwellian distribution is to lower the ionization coefficient S below the value calculated with $f^0 = f_M$. This results from the fact that the high-energy "tail" of the distribution function is depressed below its Maxwellian value. In the region of net recombination, on the other hand, for relatively large n_e/n_1 the value of S/S_M can be larger than unity. In such cases the relative overpopulation of n_2 $[n_2/n_1 > (n_2/n_1)^*]$ associated with the recombination

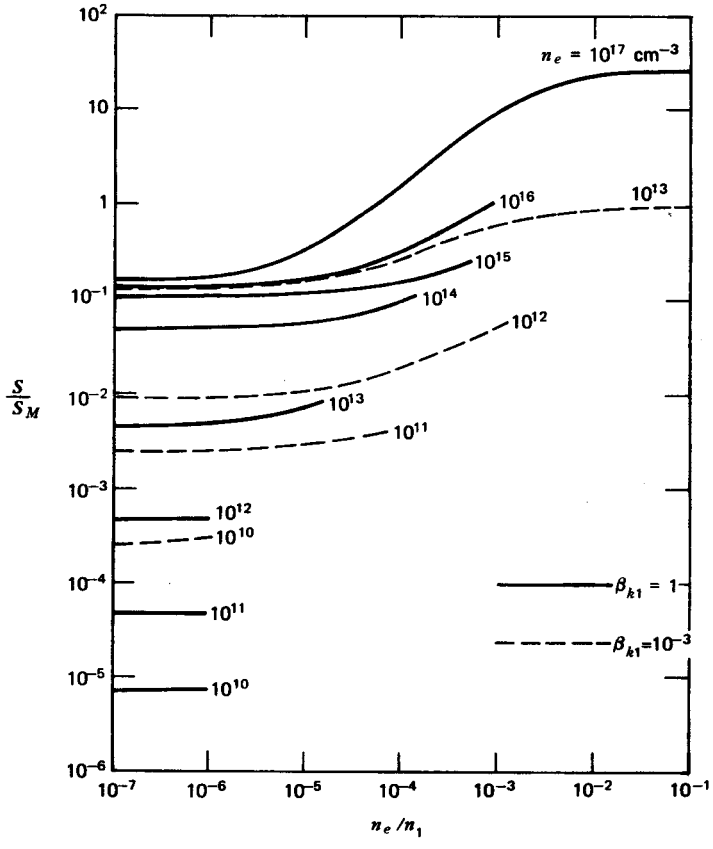


Figure 11. The effect of a non-Maxwellian distribution function on the ionization coefficient S of equation (5.18), for hydrogen at 8000°K.

process causes f^0 to be actually greater than f_M for $\epsilon > \epsilon_{12}$. This in turn increases the coefficient S.

Exercise 5.1. Find the modified form for the coefficients $a(\epsilon)$ and $b(\epsilon)$ in equation (5.6) which would account for the effects of electron-atom elastic collisions [see equation (VII 6.26)].

Exercise 5.2. Use the detailed-balancing relation (5.3) between the cross sections $Q^{1\to 2}(\epsilon)$ and $Q^{2\to 1}(\epsilon - \epsilon_{12})$ to derive for $f^0 = f_M$ the relation (2.6) between the rate integrals ${}_1S^2$ and ${}^2S_1$.

Exercise 5.3. Obtain the modified form of equation (5.6) which accounts for the nonelastic effects at low energies associated with the electron after

an inelastic collision ($e + A_1 \rightarrow e + A_2$) and the electron before a superelastic collision ($e + A_2 \rightarrow e + A_1$). By means of an order-of-magnitude argument can you estimate the conditions under which these effects can be neglected?

Exercise 5.4. Show for the solution (5.13) of the approximate Boltzmann equation (5.11) that f^0 will be greater than f_M for high-energy electrons if n_2/n_1 is greater than $(n_2/n_1)^*$. On the basis of an approximate two-level model [see equation (3.1)], explain how this situation might occur in a recombining plasma.

Exercise 5.5. Verify that with the non-Maxwellian distribution (5.13) the rate integral ${}_1S^2$ is given by equation (5.17).

Exercise 5.6. From the results given in this and the preceding section, plot the quantity $n_e/\dot{n}_e$, which is a characteristic time for ionizational relaxation, as a function of n_e for hydrogen at 8000°K and, as a parameter, $n_1 = 10^{14}$, 10^{16}, and 10^{18} cm^{-3}. For both $\beta_{k1} = 0$ and $\beta_{k1} = 1$ show curves based (*a*) on the assumption that $f^0 = f_M$, and (*b*) from the solutions of Shaw, et al., with this assumption relaxed.

6. IONIZATIONAL NONEQUILIBRIUM IN FLOWING PLASMAS

In the present section, the theory developed earlier in this chapter is brought to bear on the problem of ionizational nonequilibrium in a flowing plasma. Such a nonequilibrium situation results from the fact that a certain finite time is required for the degree of ionization of the plasma to adjust from one level to another. For a flowing plasma, the pertinent time is that measured by an observer moving with the local flow velocity. This form of ionizational nonequilibrium can result from a rapid change in an electric field or the rapid expansion of a plasma through a supersonic nozzle. It is to be contrasted to the ionizational nonequilibrium at steady state associated with radiation escape, which was discussed in Sec. 3. The effect of radiation escape, however, may also be important in a flowing plasma and, as we saw in Sec. 4, can affect the recombination and ionization rate, as well as the local "steady-state" value of the degree of ionization. Ionizational nonequilibrium as a result of rapid changes in a flow is similar to vibrational and chemical nonequilibrium in a high-temperature gas. Accordingly, much of the discussion of such gas flows, as given, for example, by Vincenti and Kruger (1965), Chapter VIII, and by Clarke and McChesney (1964), Chapter 6, has a more or less direct counterpart in problems of ionizational nonequilibrium. On the other hand, there are important differences, particularly with respect to the close

coupling of the electron energy equation and the recombination and ionization rates.

In this section, we shall consider ionizational processes in the steady flow of a partially ionized plasma in a duct. Here "steady" refers to a flow that is locally time-independent ($\partial/\partial t = 0$) at each point in the flow and, as is explained more fully later, has a different meaning from our use of "steady state." This duct can consist of the actual physical boundaries to the flow or can be any mathematical surface that is locally parallel to the mean-velocity vector, thus forming a "streamtube." For simplicity we shall adopt the quasi-one-dimensional approximation of gas dynamics, which is discussed by Shapiro (1953), Chapter 4, and Liepmann and Roshko (1957), Chapter 2. That is, with the duct axis aligned in, say, the x-direction, we regard n_e, T_e, and the other quantities of interest as uniform on any plane perpendicular to the x-direction and hence as functions of x only. In addition, we shall restrict our attention to situations where the electron current density and hence diffusion velocity are zero.

For quasi-one-dimensional flow the pertinent form of the electron continuity equation (4.4) can be derived from the consideration of a "control volume" of cross-section area equal to the area $A(x)$ of the duct and of length dx in the direction of flow (see Fig. 12). The rates at which electrons are carried by the flow into and out of this volume are

$$n_e u A \tag{6.1a}$$

and

$$n_e u A + \frac{d}{dx}(n_e u A)\, dx, \tag{6.1b}$$

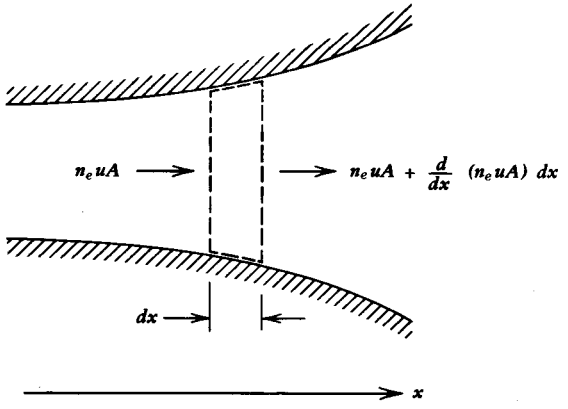


Figure 12. Control volume for the one-dimensional electron continuity equation.

respectively. Here we denote by u the component in the x-direction of the mean mass velocity $\mathbf{u}$. In the one-dimensional formulation, since the flow is in the x-direction, we can think of $u(x)$ as the only nonzero component of $\mathbf{u}$. Insofar as the area A varies with x, however, for purely geometric reasons this cannot be precisely correct. Nevertheless, the flow rate across a surface that is perpendicular to the x-direction depends only on u, so that the expressions (6.1) would be exact if $n_e uA$ were replaced by $\int n_e u\, dA$. Since we have assumed that $\mathbf{J}_e$ is zero, there is no net flow of electrons across the walls of the duct itself. For steady flow the difference between the outflow (6.1b) and the inflow (6.1a) must then be equal to the product of the volume $A\, dx$ and the net rate $\dot{n}_e$ per unit volume of production of electrons. Thus the one-dimensional electron continuity equation is

$$\frac{d}{dx}(n_e uA) = \dot{n}_e A. \tag{6.2}$$

This same result could have been obtained from the integration of the continuity equation (4.4) over the control volume of Fig. 12, with use of the divergence theorem and the conditions $\partial/\partial t = \mathbf{U}_e = 0$, $n_e = n_e(x)$, $u = u(x)$, and $\dot{n}_e = \dot{n}_e(x)$. The production rate $\dot{n}_e$ in equation (6.2) has been discussed in Secs. 4 and 5. To keep matters as simple as possible, we shall assume in that which follows that the electron density is sufficiently large that the electron distribution function f^0 is Maxwellian and hence that the analysis of Sec. 4 is directly applicable.

Since the production rate $\dot{n}_e$ depends on the electron temperature as well as n_e, we must also consider the electron energy equation. The safest way to obtain the applicable form of the electron energy equation is to integrate equation (VIII 8.5) over the control volume of Fig. 12. If we assume conditions are such that the electron heat flux $\mathbf{q}_e$ can be neglected, this yields, with the use of the divergence theorem, $\partial/\partial t = \mathbf{J}_e = 0$, and $p_e = p_e(x)$ (see Exercise 6.1),

$$\frac{1}{A}\frac{d}{dx}\left[n_e uA\left(\frac{5}{2}kT_e + \epsilon_i\right)\right] - u\frac{d}{dx}(n_e kT_e) = n_e \frac{3}{2}k(T_h - T_e)\sum_h 2\frac{m_e}{m_h}\bar{\nu}_{eh} - \dot{R}. \tag{6.3a}$$

Here, as in equation (VIII 8.4), the net rate $\dot{N}$ per unit volume of energy loss by the free electrons in nonelastic collisions is expressed as

$$\dot{N} \simeq \dot{R} + \epsilon_i \dot{n}_e = \dot{R} + \frac{\epsilon_i}{A}\frac{d}{dx}(n_e uA), \tag{6.3b}$$

where $\dot{R}$ is the local net rate per unit volume of radiant energy loss and ϵ_i is the ionization potential from the ground level. In equation (6.3) we have

assumed that the temperature T_h is the same for all heavy particles and have taken the factor $T_h - T_e$ on the right-hand side outside the summation.

The reader familiar with the control-volume equations of gas dynamics will recognize the term $n_e u(5/2)\, kT_e$ in equation (6.3) as the energy flux associated with the electron enthalpy h_e. He may, however, regard the term $-u\, d(n_e kT_e)/dx$ as an "extra" term and, perhaps, ask why there is not a term $-en_e \mathbf{u} \cdot \mathbf{E}$ on the right-hand side which results from the work done in the motion in the electrons through a field. (Although we have assumed $\mathbf{J}_e = 0$, this does not necessarily imply that $\mathbf{E} = 0$.) The explanation is as follows: The energy equation (6.3) is applied to a stationary volume, that of Fig. 12. On the other hand, the original energy equation (VIII 8.5) from which equation (6.3) was derived is based on the form (VII 3.5) of the Boltzmann equation, which is written in terms of electron velocity $\mathbf{C}$ measured with respect to $\mathbf{u}$. Now, these equations are all correct, but the first term on the right-hand side of equation (6.3) is *not* the rate of energy gain of the electrons in elastic collisions as seen by a stationary observer. In the notation of equation (VII 3.7b) this last quantity is $\sum_h \Delta_{eh}[\frac{1}{2}m_e c^2]$. Instead, the term in equation (6.3) is $\Sigma_h \Delta_{eh}[\frac{1}{2}m_e C^2]$, the energy gain seen in the moving coordinate system. It is easy to show that

$$\Delta_{eh}[\tfrac{1}{2}m_e c^2] = \Delta_{eh}[\tfrac{1}{2}m_e C^2] + \mathbf{u} \cdot \Delta_{eh}[m_e \mathbf{C}]. \tag{6.4}$$

For the present conditions the form of the electron momentum equation consistent with the present assumptions is [cf. equation (VII 3.10b) where we neglect the term $\rho_e D\mathbf{u}/Dt$ and hence assume in equation (6.7), which follows, that $\frac{5}{2}kT_e \gg m_e u^2/2$]

$$\nabla p_e + en_e \mathbf{E}' = \sum_h \Delta_{eh}[m_e \mathbf{C}]. \tag{6.5}$$

Combining equations (6.4) and (6.5), we have

$$\sum_h \Delta_{eh}[\tfrac{1}{2}m_e C^2] - \sum_h \Delta_{eh}[\tfrac{1}{2}m_e c^2] = -\mathbf{u} \cdot \nabla p_e - en_e \mathbf{u} \cdot \mathbf{E}'. \tag{6.6}$$

Thus the "extra" term and the "missing" term in equation (6.3) reflect the difference between the elastic collisional energy gain as seen in the two reference frames. With the use of equations (6.6) and $p_e = n_e kT_e$, the energy equation (6.3) can be written in the form

$$\frac{1}{A}\frac{d}{dx}\left[n_e uA\left(\frac{5}{2}kT_e + \epsilon_i\right)\right] = -en_e \mathbf{u} \cdot \mathbf{E} + \sum_h \Delta_{eh}\left[\frac{1}{2}m_e c^2\right] - \dot{R}. \tag{6.7}$$

Although this last equation is more directly amenable to physical interpretation, from the point of view of the stationary control volume of Fig. 12, because of the term $\Delta_{eh}[\frac{1}{2}\, m_e c^2]$ it is not a form suitable for analysis.

Equations (6.2) and (6.3) describe the change in n_e and T_e in a flowing plasma. Before these equations can be put to use, explicit forms must be specified for $\dot{n}_e$ and $\dot{R}$. For simplicity we shall assume that the radiation-escape term $\dot{R}$ can be neglected in the electron energy equation (6.3) and that the production rate $\dot{n}_e$ in the electron continuity equation (6.2) is given by the Hinnov–Hirschberg formula (4.20) as

$$\dot{n}_e = \alpha[n_1(n_e^2/n_1)^* - n_e^2], \tag{6.8a}$$

where, with T_e in °K,

$$\alpha = 3.4 \times 10^{-22}\left(\frac{T_e}{1000}\right)^{-9/2} n_e \left[\frac{\text{cm}^3}{\text{sec}}\right]. \tag{6.8b}$$

Furthermore, we shall assume that u, T_h, and n_1 are known as functions of x from a solution of the conservation equations of the gas as a whole. In general, such a solution is coupled to a solution of the electron energy equation (6.3). This results from the transfer of energy between the electrons and the heavy particles exhibited by the first term on the right-hand side of equation (6.3). In certain cases, however, when the degree of ionization of the gas as a whole is not too large, this energy transfer can be neglected in the energy equation of the gas and the state of the heavy particles can be found by means of the classical methods discussed by Shapiro (1953) and Liepmann and Roshko (1957). In this case, the heavy particles can be regarded as a known "heat bath" in which the electrons are immersed.

To fix ideas, we shall first consider a situation in which the area A and also u, T_h, and n_1 are all constant. At some point in the flow, say $x = 0$, we specify that the electron density $n_e^0 = n_e(x = 0)$ is in Saha equilibrium with the electron temperature $T_e^0 = T_e(x = 0)$ and that T_e^0 is greater than T_h. This situation, in which T_e^0 is elevated above T_h and n_e^0 is given by the Saha equation evaluated at T_e^0, could result from the $\mathbf{J}_e \cdot \mathbf{E}$ ohmic heating of the electrons by a discharge in the region $x < 0$ (see, *e.g.*, Fig. 2 in Chapter VIII, Sec. 3). We then ask, for the recombination process in the region $x > 0$ where $\mathbf{J}_e$ is now zero, how T_e and n_e adjust to their final equilibrium values $T_e^f = T_h$ and $n_e^f = n_e^*(T_e^f = T_h)$. In this situation the electron continuity and energy equations (6.2) and (6.3) in the absence of radiation escape reduce to

$$u\frac{dn_e}{dx} = \alpha\left[n_1\left(\frac{n_e^2}{n_1}\right)^* - n_e^2\right] \tag{6.9}$$

and

$$u\frac{d}{dx}\left[n_e\left(\frac{3}{2}kT_e + \epsilon_i\right)\right] = -n_e\frac{3}{2}k(T_e - T_h)\sum_h 2\frac{m_e}{m_h}\bar{\nu}_{eh}, \tag{6.10}$$

where α is given in terms of T_e by equation (6.8b). The initial conditions are

$$\text{at } x = 0 \begin{cases} T_e = T_e^0 > T_h \\ n_e = n_e^0 = [n_1(n_e^2/n_1)^*(T_e = T_e^0)]^{1/2} \end{cases}. \tag{6.11}$$

Equations (6.9) through (6.11) describe two coupled forms of nonequilibrium: the elevation of T_e above T_h and the elevation of n_e above its final Saha value n_e^f. Both forms of nonequilibrium result from the finite "time," measured here as x/u, for the recombining plasma to adjust from the initial conditions T_e^0, n_e^0 to the final conditions T_e^f, n_e^f. This situation is equivalent to the analogous recombination problem in a transient stationary plasma for which at time $t = 0$, $T_e = T_e^0$ and $n_e = n_e^0$. The equations for this case are obtained from equations (6.9) and (6.10) by means of the transformation $t = x/u$.

For given values of the parameters it is not too difficult to solve the simultaneous equations (6.10) and (6.11) numerically for $T_e(x)$ and $n_e(x)$. We shall consider only two special cases: First, suppose the recombination rate given by equations (6.8) is so rapid in comparison with the rate of adjustment of T_e that, for each x, n_e is nearly in Saha equilibrium with the local value of T_e. Then locally we have

$$n_e^2 \simeq (n_e^*)^2 = n_1 \left(\frac{2\pi m_e kT_e}{h^2}\right)^{3/2} \frac{2g_i}{g_1} e^{-\epsilon_i/kT_e}. \tag{6.12}$$

If we use this form for n_e in the energy equation (6.10) and neglect terms of order kT_e/ϵ_i, we then obtain

$$\frac{1}{2}\left(\frac{\epsilon_i}{kT_e}\right)^2 u \frac{dT_e}{dx} = -\frac{3}{2}(T_e - T_h) \sum_h 2 \frac{m_e}{m_h} \bar{\nu}_{eh}. \tag{6.13}$$

In the present example, although we assume that as a result of a rapid recombination rate n_e is nearly equal to n_e^*, as indicated by equation (6.12), this does *not* mean that the continuity equation (6.9) then reduces to

$$u \frac{dn_e}{dx} = 0.$$

This form of the continuity equation for constant u and A applies only in the opposite, *frozen-flow* limit of slow recombination, where $\alpha \to 0$. In the present, *quasiequilibrium* limit the recombination coefficient α is regarded as large, so that the continuity equation (6.9) is satisfied by $\alpha \to \infty$ and $n_e^2 \to n_e^{*2}$ in such a way that

$$u \frac{dn_e}{dx} \to u \frac{dn_e^*}{dx}.$$

To the extent that $\bar{\nu}_{eh}$ can be taken as approximately constant and independent of T_e, the solution to equation (6.13), with the initial condition (6.1), is

$$\left(\frac{T_e - T_h}{T_e^0 - T_h}\right)\frac{T_e^0}{T_e} \times \exp\left\{\frac{T_h}{T_e} - \frac{T_h}{T_e^0}\right\} = \exp[-(x/u)/\tau_T']. \tag{6.14}$$

Here τ_T' is for this problem a characteristic relaxation time for the electron temperature given by

$$\tau_T' = \frac{(\epsilon_i/kT_h)^2}{6\sum_h (m_e/m_h)\bar{\nu}_{eh}}. \tag{6.15}$$

Although the form of the solution (6.14) is complicated as a result of the factor T_e^{-2} on the left-hand side of equation (6.13), the qualitative behavior of the solution is that of exponential relaxation with a characteristic time τ_T', at least as long as T_e^0/T_h is not too large (see Exercise 6.5).

We now consider the opposite situation in which the relaxation time for T_e is taken to be small when compared with the relaxation time for n_e. That is, we assume that T_e is given by its local "steady-state" value which is determined by a balance between the rate $\epsilon_i|\dot{n}_e|$ of energy gain as a result of nonelastic collisions in the recombination process and the rate of energy loss in elastic collisions. When this is the case, T_e can be found in terms of n_e from the relation

$$-\epsilon_i \dot{n}_e = \epsilon_i \alpha[n_e^2 - n_1(n_e^2/n_1)^*] = n_e \frac{3}{2} k(T_e - T_h)\sum_h 2\frac{m_e}{m_h}\bar{\nu}_{eh}. \tag{6.16}$$

Here, as previously, we use "steady state" in the sense that there exists a local balance between gain and loss mechanisms. This is, of course, different from steady flow ($\partial/\partial t = 0$). Furthermore, the local steady state is not necessarily an equilibrium or final state. In a local steady-state solution such as that of equation (6.16), the terms that involve convective spatial derivatives, and in general the time derivative, are neglected in the pertinent equation. This does not mean that spatial changes are ignored, but rather that the other terms in the equation are regarded as of dominant magnitude. Thus, equation (6.16) is obtained from equations (6.9) and (6.10) if in equation (6.10) the term $ud(n_e \frac{3}{2}kT_e)/dx$ is neglected in comparison with the other terms. We note that it would not be appropriate in this context to neglect the term $u\epsilon_i\, dn_e/dx$ in equation (6.10), since physically this term accounts for the rate of energy gain in nonelastic collisions.

Given the implicit dependence of T_e on n_e contained in equation (6.16) it is possible to integrate numerically the electron continuity equation (6.9) to obtain $n_e(x)$. To keep matters as simple as possible, we shall be content with the solution for which T_e and hence $(n_e^2/n_1)^*$ and α/n_e are taken to be constant. Physically this would correspond to the case where the elevation of T_e above T_h given by equation (6.16) is everywhere small. The solution to equation (6.9) is then

$$\frac{1-(n_e^f/n_e)^2}{1-(n_e^f/n_e^0)^2} = e^{-(2x/u)/\tau_n}, \tag{6.17}$$

where

$$(n_e^f)^2 = n_1(n_e^2/n_1)^*$$

and where a characteristic recombination time τ_n is

$$\tau_n = \frac{1}{(n_e^f)^2\alpha/n_e}. \tag{6.18}$$

Since by equation (6.8b) α/n_e is independent of n_e, τ_n is a function only of T_e. (We assume that $n_1 \gg n_e$, so that $n_1 \neq n_1(n_e)$.)

In the process that we have just considered, we have assumed that, in a certain sense, the relaxation of the electron temperature is much faster than the relaxation of n_e. It might be assumed that this will be the case if τ_T' from equation (6.15) is much less than τ_n from equation (6.18). In fact, this is *not* the case. The appropriate condition is instead

$$\tau_T \ll \tau_n,$$

where

$$\tau_T = \frac{1}{\sum_h 2(m_e/m_h)\bar{\nu}_{eh}}. \tag{6.19}$$

This can be seen from the following argument: At $x = 0$, where $T_e = T_e^0$ and $n_e = n_e^0 = n_e^*(T_e^0)$, the continuity equation (6.9) yields $dn_e/dx = 0$. Therefore, from the energy equation (6.10), we have at $x = 0$

$$\frac{u\,dT_e}{dx} = \frac{-(T_e^0 - T_h)}{\tau_T},$$

so that at least at the initial point the characteristic time is τ_T. Furthermore, if we assume that T_e relaxes over a distance, say $0 > x > x'$, in which n_e is approximately constant but n_e^* decreases significantly as a

result of the decrease in T_e, then for this distance the original energy equation (6.10) can be written as

$$u\frac{d}{dx}\left(\frac{3}{2}kT_e\right) = \frac{-\frac{3}{2}k(T_e - T_h)}{\tau_T} - \left(\frac{3}{2}kT_e + \epsilon_i\right)\dot{n}_e/n_e$$

$$\simeq \frac{-\frac{3}{2}k(T_e - T_h)}{\tau_T} + \epsilon_i\frac{\alpha}{n_e}(n_e^2 - n_e^{*2})$$

$$\leq \frac{-\frac{3}{2}k(T_e - T_h)}{\tau_T} + \left[\frac{\epsilon_i\alpha}{n_e}(n_e^2 - n_e^{*2})\right]_{x=x'},$$

where $n_e^{*2} \equiv n_1(n_e^2/n_1)^*$. If we now denote by T_e' the value of T_e at $x = x'$ [where equation (6.16) is satisfied], the last form of the energy equation can be written as

$$u\frac{d}{dx}\left(\frac{3}{2}kT_e\right) \leq \frac{-\frac{3}{2}k(T_e - T_e')}{\tau_T},$$

with the approximate solution

$$\frac{T_e - T_e'}{T_e^0 - T_e'} \leq \exp[-(x/u)/\tau_T].$$

Thus, for the present situation of relatively slow variation of n_e the initial relaxation of T_e to the value given by equation (6.16) is at least as fast as that characterized by τ_T.

The appearance of two different relaxation times τ_T and τ_T' in our two examples may seem to the reader complicated at best. These examples, however, illustrate the important consideration that the safest practice is to estimate the relaxation times separately for each physical situation directly from the governing equations. For many plasma conditions, it is found that the relaxation time τ_T for T_e is less than the relaxation time τ_n for n_e, so that the adjustment of T_e to local conditions is more rapid than that of n_e (see Exercise 6.6).

We now turn our attention to a somewhat more complicated problem, that of the expansion of a partially ionized plasma through a supersonic nozzle. We assume that the plasma is initially under equilibrium conditions upstream of the nozzle where the velocity is low, so that there, in the absence of radiation escape, the electron and heavy-particle temperatures are equal and the electron density is given by the Saha equation. During the expansion the heavy-particle temperature will decrease as the result of the transformation of thermal energy to directed energy of mean motion. Elastic collisions between electrons and heavy particles will then tend to lower the

electron temperature, which in turn will cause the plasma to recombine. The change in flow area and the decrease in ground-level density will, to usually a lesser extent, also act to lower the electron density. At a sufficient distance downstream of the nozzle in the supersonic region, a new equilibrium state will be reached at a lower temperature and correspondingly lower electron density as compared to the upstream state.

Between the initial and final conditions, both the electron temperature and the electron density may depart from their equilibrium values. Since in the process of recombination energy is gained by the free electrons at the rate $-\epsilon_i \dot{n}_e$, the electron temperature will tend to be elevated above the heavy-particle temperature. Thus, even when the gradients occurring in the expansion are sufficiently small that local "steady-state" conditions prevail, $T_e - T_h$ must still satisfy equation (6.16). Furthermore, for typical experimental conditions the characteristic time for changes in the flow $[\sim (u\, d/dx)^{-1}]$ will be smaller in certain regions of the expansion than the characteristic times for adjustment of T_e and n_e. These latter times can be estimated from equations (6.18) and (6.19) or (6.15), depending on the circumstances. In particular, the recombination rate, which is proportional to n_e^3, will be low in the downstream, supersonic region of the nozzle where n_e is relatively low [see equations (6.8)]. As a result, the electron density in this region cannot fall much below the value n_e^f for which the recombination time $(n_e/\alpha)(n_e^f)^{-2}$ is equal to the flow time. Of course, the flow time required to reach a point far downstream of the nozzle is arbitrarily long, and n_e will eventually decrease to its final equilibrium value.

The nonequilibrium expansion through the nozzle, for the case of zero electron current density and heat flux and one-dimensional flow, is described by the continuity and energy equations (6.2) and (6.3). In these equations, $A(x)$ is presumably known and, if the elastic energy transfer from the electrons can be neglected in the heavy-particle energy equation, the velocity and properties of the heavy species can be separately calculated from the conservation equations of the gas as a whole. When $kT_e \ll \epsilon_i$, the ground-level density n_1, which enters in the evaluation of $\dot{n}_e$ [see equation (6.8a)] is to a good approximation independent of the degree of ionization and can also be calculated separately.

As a result of the complexity of the complete formulation given by equations (6.2) and (6.3), it is desirable to seek somewhat simpler approximate solutions. One such approach is the prescription of "sudden freezing" suggested for problems of chemical nonequilibrium by Bray (1959). In this approximation, as applied here to the electron continuity equation (6.2), it is assumed that n_e is equal to the Saha-equilibrium value

$$n_e^*(T_e) = [n_1(n_e^2/n_1)^*]^{1/2} \tag{6.20}$$

in the upstream portion of the nozzle where the kinetic recombination rate $\alpha^* n_e^{*2} A$ is greater than the recombination rate $-d(n_e^* uA)/dx$ demanded by the flow. Beyond the position where

$$-\frac{d}{dx}(n_e^* uA) = \alpha^* n_e^{*2} A, \tag{6.21}$$

the electron density is then regarded as "frozen." That is, the calculation is continued with $\dot{n}_e$ set equal to zero in the continuity equation (6.2), which then becomes

$$\frac{d}{dx}(n_e uA) \simeq 0. \tag{6.22}$$

In such a frozen flow, n_e will change as a result of the change with x of the product uA. The relative electron concentration n_e/n_h, however, will remain constant. (This is true in the absence of chemical reactions between the heavy species; in general it is ρ_e/ρ that will be constant.)

As an extension of the sudden-freezing method, the electron continuity equation beyond the position where equation (6.21) is satisfied can be written as

$$\frac{d}{dx}(n_e uA) \simeq -\alpha n_e^2 A \tag{6.23}$$

rather than as equation (6.22). The argument here is that beyond the position specified by equation (6.21) as a result of the decrease in T_e the ionization rate $\alpha n_1 (n_e^2/n_1)^* A$ will tend to be small in magnitude relative to the recombination rate $\alpha n_e^2 A$. Accordingly, the ionization term is neglected in the approximate continuity equation (6.23). In a recombining plasma the net recombination rate calculated on the basis of this model will tend to be greater than that given by the complete equations, so that locally n_e will tend to be lower and T_e higher than the exact values.

Calculations based on the foregoing approximation have been performed by Bencze (1965) for expansion of 0.25 mole percent cesium- and potassium-seeded argon through a supersonic nozzle of overall length about 15 cm from an initial stagnation temperature of 2000°K and an initial stagnation pressure of 5 atm. In the region upstream of the point where equation (6.21) is satisfied n_e was assumed equal to $n_e^*(T_e)$ as given by equation (6.20) and $T_e(x)$ was calculated from the electron energy equation [cf. equation (6.13)]. Downstream of the position specified by equation (6.21), $T_e(x)$ and $n_e(x)$ were found from the simultaneous solution of the energy equation and the approximate continuity equation (6.23). The energy equation employed was of the form (6.3), with radiation escape neglected and the term $-u\, d(n_e kT_e)/dx$ omitted. For the cases considered this omission probably had little effect on the results. The Hinnov and Hirschberg

formula (6.8) was used for $\dot{n}_e$ and the expansion of the heavy species was assumed to be independent of the recombination process.

The results of these calculations by Bencze are shown in Fig. 13 for the case of potassium-seeded argon. In addition to the calculated values of n_e and T_e, curves of n_e are shown for completely frozen and completely equilibrium conditions. In the former case, $\dot{n}_e$ is zero everywhere and, since $n_e uA$ is then constant, n_e decreases only because of the increase in the product uA. The complete-equilibrium case corresponds to Saha equilibrium at the local heavy-particle temperature. In addition there is a curve for n_e corresponding to the sudden-freezing approximation [see equation (6.22)]. The position at which equation (6.21) is satisfied can be identified by the symbol *.

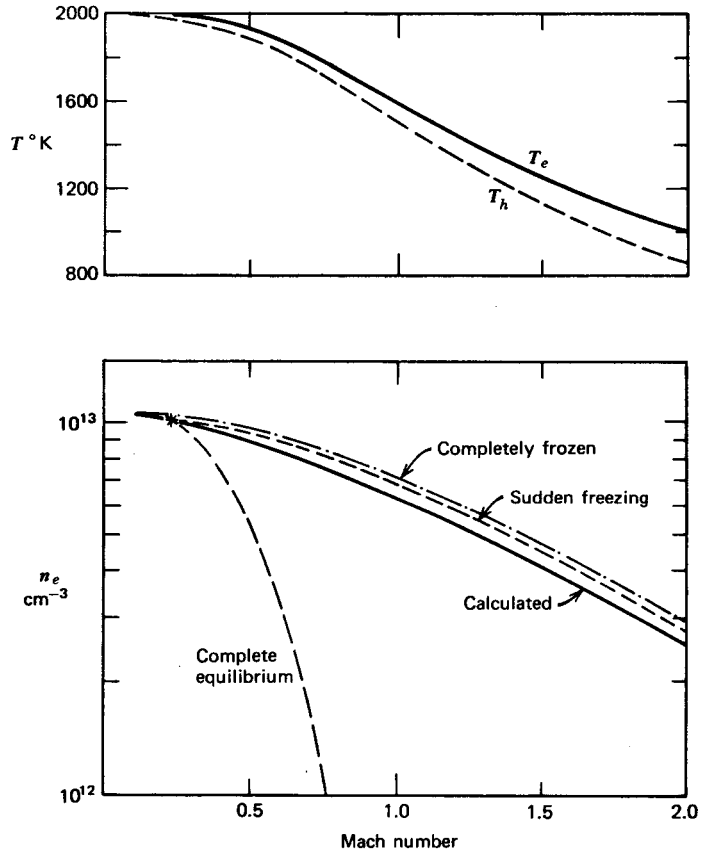


Figure 13. The results of the nozzle-flow calculations of Bencze (1965) for potassium-seeded argon.

Exercise 6.1. Derive the control-volume continuity and energy equations (6.2) and (6.3) by integration of the corresponding partial-differential equations (4.4) and (VIII 8.5) over the control volume of Fig. 12.

Exercise 6.2. Show that the approximation indicated in equation (6.3b) is equivalent to the approximation (4.1) of Sec. 4 that $\dot{n}_{k>1} = 0$.

Exercise 6.3. Give a physical interpretation of the various terms in the electron energy equation in the form (6.7).

Exercise 6.4. Derive the form (6.13) of the electron energy equation that applies for constant-area flow with $n_e = n_e^*(T_e)$.

Exercise 6.5. From the solution (6.14) of the electron energy equation, plot $(T_e - T_h)/(T_e^0 - T_h)$ as a function of $x/u\tau_T'$ for the case where $T_e^0/T_h = 2$.

Exercise 6.6. Estimate values for the relaxation times τ_n, τ_T, and τ_T' given by equations (6.18), (6.19), and (6.15) for argon and 0.25 mole percent potassium-seeded argon plasmas at several values of T_e, including those of Table 1, Sec. 3. For simplicity, use the approximate constant cross sections $Q_{eA} \sim 10^{-16}$ cm^2 and $Q_{ek} \sim 4 \times 10^{-14}$ cm^2 and estimate $\tau_n = 1/\alpha n_e$ with the use of the Saha equation. The ratios τ_T/τ_n and τ_T'/τ_n at a given T_e will then be approximately independent of the plasma pressure. Under what conditions is the relaxation time for T_e smaller than the relaxation time for n_e?

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